

ECMA 30760 (S26)
Introduction to Economic Design

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1 Overview

Big picture: how to design economic institutions.

Example 1.1. Voting. Auctions. Matching (think marriage or school choice). Resource allocation (think generalization of auctions). 

Our role:

- Sometimes, we're going to imagine ourselves as a benevolent planner.
- Other times, we'll try to extract surplus.

Challenges:

- People are strategic. (Game theory)

Example 1.2 (Boston public school, failed to account for strategic behavior). Everyone report ranked preferences of schools. 

- Information and power are decentralized.

Methods: Axioms (mathematical).

Techniques: Mostly basic calculus, simple logic, linear programming, linear algebra.

2 Voting

2.1 Setup

- Finite set A of outcomes.
- Finite set N of agents/voters.
- Each agent i has a **strict preference** $\succ_i$ over A . We write $x \succ_i y$ if agent i strictly prefers outcome x to outcome y , and $x \succeq_i y$ if either $x \succ_i y$ or $x = y$.
- Let P be the set of strict orders on A . Let P^N be the set of all preference profiles, i.e., $P^N = \{(\succ_1, \dots, \succ_n) : \succ_i \in P, \forall i\}$.
- A **social choice function** is a mapping f from P^N into A .

2.1.1 Preferences

Definition 2.1. A **strict preference** $\succ_i$ is a **binary relation**, i.e., a subset of $A \times A$, with the following properties:¹

- irreflexive: for all $x \in A$, $x \not\succeq_i x$.
- complete: for each $x \neq y$, either $x \succ_i y$ or $y \succ_i x$.
- transitive: if $x \succ_i y$ and $y \succ_i z$, then $x \succ_i z$.
- asymmetric: for each $x \neq y$ if $x \succ_i y$ then we don't have $y \succ_i x$.

Such binary relations are also called **linear orders**. 

Note that $x \not\succeq_i y$ implies $x \preceq_i y$, i.e., $y \succ_i x$ or $x = y$. But since strict orders can be embedded into weak orders, we can also write $x \preceq_i y$.

2.1.2 Social choice functions

Example 2.2 (Voting rules).

- **Plurality/First-past-the-post** selects the outcome that is ranked first by the most voters, with some tie breaking rule.
- **Instant Runoff** selects the two highest ranked outcomes, and the one that is ranked higher by more voters is selected.
- **Elimination** repeatedly eliminates the outcome that is ranked highest by the fewest voters, until only one outcome remains.
- **Borda count**: Let $n := |A|$ be the number of outcomes. Give n points to a candidate for each agent that top-ranks them; $n - 1$ points for each agent that ranks them second; and so on. The candidate with the most points wins. 

Example 2.3 (Outcomes). $A = \{x, y, z, w\}$. Preferences are as follows:

- 6 people: $w \succ_i x \succ_i y \succ_i z$.
- 5 people: $x \succ_i y \succ_i w \succ_i z$.
- 4 people: $y \succ_i x \succ_i w \succ_i z$.
- 3 people: $z \succ_i y \succ_i x \succ_i w$.

¹We use the **infix** notation, writing $x \succ_i y$ instead of $(x, y) \in \succ_i$.

Plurality selects w . Instant runoff selects x . Elimination selects y . Borda count selects x (w, x, y, z get resp., 45, 56, 52, 27 points). 

Borda count is a special case of a

Definition 2.4 (Summation social choice function). We assign weights $\varphi_i(m)$ for agent i 's m th ranked outcome. 

This kind of social choice function is not popular in elections, but is popular in sports.

Another class of social choice functions is the

Definition 2.5 (Pairwise Comparison Graph). For each pair of candidates, we ask who would win in a head-to-head election and by what margin and plot the results in a directed graph, with weights the margins of victory and arrow pointing from the winner to the loser.² The **Copland** methods counts for each outcome how many arrows pointing out and choose whichever has most. 

Copland selects y in the example above.

2.2 Properties of Social Choice Functions

Definition 2.6. An outcome x is a **Condorcet winner** if it beats every other outcome in a head-to-head election. We say a social choice function satisfies the **Condorcet property** if it selects the Condorcet winner whenever one exists. 

In the example above, a Condorcet winner is x . However, a Condorcet winner does not always exist:

Example 2.7.

$$\begin{aligned} x &>_i y >_i z \\ y &>_i z >_i x \\ z &>_i x >_i y \end{aligned}$$



Definition 2.8. A set of outcomes $A' \subset A$ is called **dominating** if it is nonempty and for each $x \in A', y \in A \setminus A', x$ beats y in a head-to-head election. 

Lemma 2.9. If X and Y are dominating sets, then $X \subset Y$ or $Y \subset X$.

Proof. If not, any $x \in X \setminus Y$ and $y \in Y \setminus X$ have to beat each other in a head-to-head election, a contradiction. $\square$

Thus we can define a smallest dominating set:

Definition 2.10. The **Smith set** is the intersection of all dominating sets. We say f satisfies the **Smith property** if it always selects an outcome in the Smith set. 

²I.e., the number of votes for the winner minus the number of votes for the loser.

This generalizes the Condorcet property: if a Condorcet winner exists, it is the only element of the Smith set.

Definition 2.11. A social choice function f is **strategy-proof** if there is no preference profile $(\succ_1, \dots, \succ_n)$, agent i , and preference $\succ'_i$ such that

$$f(\succ_1, \dots, \succ_{i-1}, \succ'_i, \succ_{i+1}, \dots, \succ_n) \succ_i f(\succ_1, \dots, \succ_n). \quad \text{👉}$$

Note that this does not rule out collusion being profitable.

Exercise 2.12. Show that if $|A| > 2$, then none of the social choice functions introduced so far are strategy-proof.

2.3 The Gibbard-Satterthwaite Theorem

Theorem 2.13 (Gibbard-Satterthwaite). *If $|A| \geq 3$, then every surjective and strategy-proof social choice function is **dictatorial**, i.e., there exists an agent i such that $f(\succ) = \arg \max_{A} \succ_i$ for each preference profile $\succ$.*

Proof outline: (1) Derive two properties of surjectivity and strategy-proofness: monotonicity and Pareto efficiency. (2) Assume $N = 2$. (3) Induct.

Definition 2.14. Given $\succ_i$ on A and $a \in A$, write the lower contour set of a as $L_{\succ_i}(a) := \{b : a \succ_i b\}$. 👉

Definition 2.15. We say f is **monotonic** if for each pair of profiles $\succ, \succ'$ such that

$$L_{\succ'_i}(f(\succ)) \supset L_{\succ_i}(f(\succ)), \quad \forall i,$$

we have $f(\succ') = f(\succ)$. 👉

In words, f is monotonic if whenever the outcome under $\succ$ is weakly better for each agent under $\succ'$, then the outcome is the same under $\succ'$.

Remark 2.16. Note that an implication of monotonicity is that the outcome does not change under permutations of outcomes ranked below/higher the outcome under each $\succ_i$. ☕

Lemma 2.17. *If f is strategy-proof, then f is monotonic.*

Proof. Let $\succ, \succ'$ be as in the definition of monotonicity, so that $f(\succ)$ weakly improves under a preference change from $\succ$ to $\succ'$. By strategy-proofness, we have

$$f(\succ) \succeq_i f(\succ'_i, \succ_{-i}) \quad \forall i.$$

If it is the case that $f(\succ) \succ_1 f(\succ'_1, \succ_{-1})$, then by definition of $\succ, \succ'$, we have $f(\succ) \succ'_1 f(\succ'_1, \succ_{-1})$, a violation of strategy-proofness. Thus $f(\succ) = f(\succ'_1, \succ_{-1})$. We can repeat this argument to conclude

$$f(\succ) = f(\succ'_1, \succ_{-1}) = f(\succ'_1, \succ'_2, \succ_{-1,2}) = \dots = f(\succ'). \quad \square$$

Definition 2.18. f is **Pareto efficient** if $a \succ_i b$ for each i implies $f(\succ) \neq b$. 

Lemma 2.19. *Monotonicity and surjectivity imply Pareto efficiency.*

Proof. Suppose for contradiction that $a \succ_i b$ for each i but $f(\succ) = b$. Construct new profile $\succ'$ by moving a to the top in each $\succ_i$. Then, $f(\succ') = b$ by monotonicity. By surjectivity, there exists $\tilde{\succ}$ such that $f(\tilde{\succ}) = a$. Modify $\tilde{\succ}$ to $\tilde{\succ}'$ by moving a to the top of all agent's list. Again, $f(\tilde{\succ}') = a$ by monotonicity. Note that both $\succ'$ and $\tilde{\succ}'$ has a at the top of each agent's list, so $f(\succ') = f(\tilde{\succ}')$, a contradiction. $\square$

We can now prove the Gibbard-Satterthwaite theorem:

Proof. The lemmas above imply that f is monotonic and Pareto efficient. We now show that f is dictatorial by inducting on n .

Base case: Consider first the case $n = 2$. Suppose

$$\begin{aligned} a &\succ_1 b \succ_1 \dots \\ b &\succ_2 a \succ_2 \dots \end{aligned}$$

By Pareto efficiency, $f(\succ) \in \{a, b\}$. Assume without loss that $f(\succ) = a$. Consider the following new profile:

$$\begin{aligned} a &\succ_1 b \succ_1 \dots \\ b &\succ'_2 \dots \succ'_2 a. \end{aligned}$$

We know that $f(\succ') \neq b$ since otherwise $f(\succ) = b$. One can also see this from strategy-proofness. It is also not any $c \neq a, b$ by Pareto efficiency. We conclude $f(\succ') = a$. By monotonicity, $f(\tilde{\succ}) = a$ whenever $\arg \max_{\tilde{\succ}_1} = a$. We may say that 1 is the dictator over a .

Now pick any $c \neq a, b$ and consider

$$\begin{aligned} c &\succ_1 b \succ \dots \\ b &\succ_2 c \succ_2 \dots, \end{aligned}$$

where we reuse the symbol $\succ_i$ for the new preferences above. Can $f(\succ) = b$ and agent 2 be the dictator over b ? No, since a contradiction arises if agent 1 and 2 respectively top ranks a and b . Thus $f(\succ) = c$ and agent 1 is the dictator over c . Since $c \neq b$ is arbitrary, agent 1 is the dictator over all outcomes except maybe b .

For b , consider³

$$\begin{aligned} b &\succ_1 a \succ \dots \\ a &\succ_2 b \succ_2 \dots \end{aligned}$$

If $f(\succ) = a$, then 2 is a dictator over a and by the same argument as above, a dictator over all outcomes except maybe b . A contradiction arises by observing that both agents are dictators over $c \neq a, b$.⁴ Thus agent 1 must also be a dictator over b .

³This is where we used $|A| > 2$.

⁴When $|A| = 2$, it is possible that both agents are also dictators over some $a \in A$. An example would be a mechanism that chooses a if and only if either agent top ranks a . Thus we examine c here.

Induction: Assume that Gibbard-Satterthwaite holds with $2, \dots, n$ agents. Let f be a surjective and strategy-proof social choice function with $n + 1$ agents.

Define

$$g(>_1, >_2) := f(>_1, >_2, >_2, \dots, >_2).$$

Note that g is surjective since f is Pareto efficient (consider the case where everyone top ranks some alternative). We claim that g is also strategy-proof. To see this, fix $>_1$ and consider $>_2$ and $>'_2$. By strategy-proofness of f , we have

$$\begin{aligned} f(>_1, >_2, >_2, \dots, >_2) &\succeq_2 f(>_1, >'_2, >_2, \dots, >_2) \\ &\succeq_2 \dots \\ &\succeq_2 f(>_1, >'_2, >'_2, \dots, >'_2) = g(>_1, >'_2). \end{aligned}$$

By transitivity, $g(>_1, >_2) \succeq_2 g(>_1, >'_2)$. Thus g is strategy-proof for agent 2. Strategy-proofness for agent 1 is inherited from f . By induction hypothesis, g is dictatorial.

If 1 is the dictator for g , we can conclude by monotonicity. We consider thus the case where 2 is the dictator of g . Fix $>^*_1$ and define

$$h(>_2, \dots, >_N) := f(>^*_1, >_2, \dots, >_N).$$

We know that h is surjective by considering the case where all agents except 1 top rank some alternative, since 2 is the dictator for g . It also inherits strategy-proofness from f . By induction hypothesis, there exists dictator k of h . It remains to argue that k is independent of the choice of $>^*_1$. For this, fix $>^*_2, \dots, >^*_{k-1}, >^*_{k+1}, \dots, >^*_N$ and define

$$q(>_1, >_k) := f(>_1, >^*_2, \dots, >^*_{k-1}, >_k, >^*_{k+1}, \dots, >^*_N).$$

Again, q inherits strategy-proofness from f . It must also be surjective: consider just $>_1 = >^*_1$, whence k becomes the dictator for h and can top rank any alternative. By the 2-agent case, k is a dictator for q . But this works for each choice of $>^*_j$, so k is the dictator for f . $\square$

2.4 Criticisms of Gibbard-Satterthwaite

Strategy-proofness is too strong. Once we fix a social choice function, it induces a game, where strategy-proofness is equivalent to seeking a dominant strategy in the game. Alternatively, we can seek a Nash equilibrium implementation, or a BNE implementation. Though note that in the latter case our results may depend on the information structure.

Domain is too large. We often have more structure on preferences in particular contexts.

Example 2.20.

- If preferences are single-peaked, then there are strategy-proof and non-dictatorial social choice functions.
- Indifferences: in allocation problems we can assume agents only care about their own allocation. 

Why not Randomize? Allowing for lotteries, there is always a random Condorcet winner. But Gibbard (1977) showed that any strategy-proof mechanism $f : P^n \rightarrow \Delta(A)$ is a randomization over mechanisms that are either dictatorial or “duple” (i.e., only select two outcomes).

Cardinal Information. One might think cardinal utility might help. It doesn't. It turns out that a strategy-proof social choice function will not use the cardinal information.

3 Aggregation

If we can know the preferences of each agent, what should our (the benevolent planner's) own preferences be?

3.1 Ordinal Aggregation

Definition 3.1. Let R be the set of weak preferences on A , i.e., allowing ties. So R is the set of complete, transitive, and reflective ($a \geq a$ for each a) binary relations on A . 

Definition 3.2. A **strict social welfare function** is a mapping $F : P^N \rightarrow P$. A **weak social welfare function** is a mapping $F : P^N \rightarrow R$. 

Example 3.3.

- Dictatorship: $F(>) = >_1$.
- Elimination method: rank in increasing order at the round they left.
- Borda count.
- Copeland.
- For any two strict rankings $>_i$ and $>'_i$ on A , define the **Kendall-Tao** distance $d(>_i, >'_i)$ to be the number of pairs $a, b \in A$ such that $a >_i b$ and $a <'_i b$ or the reverse direction for both. The **Kemeny-young** social welfare function $F(>)$ minimizes

$$\sum_{i=1}^n d(F(>), >_i).$$

- **Dodgson.** For any outcome a and any preference profiles $>$, let $>^a$ be the profile such that

- a is a Condorcet winner,
- $\sum_i d(>_i, >_i^a)$ is minimized.

Then assign $z(a) = \sum_i d(>_i, >_i^a)$, so z measures how far a is from being a Condorcet winner. Rank a above b if and only if $z(a) \leq z(b)$. 

Definition 3.4. A social welfare function F satisfies the **Paretian property** if for any $>$ and $a, b \in A$ is such that $a >_i b$ for each i , then $aF(>)b$. 

Definition 3.5. A social welfare function F satisfies **independence of irrelevant alternatives** (IIA) if for any two profiles $>, >'$ such that for each i , $>_i$ and $>'_i$ agree when restricted to $\{a, b\}$ for some $a, b \in A$, then

$$aF(>)b \iff aF(>')b.$$



Example 3.6. Suppose $a >_1 b >_1 c$ and $b >_2 c >_2 a$. By Paretian we have $b >_s c$, where the subscript s stands for social.

One might think that b should be ranked above a . IIA disallows this type of reasoning, because the relative ranking of a and b should not depend on how many other outcomes d_i we may insert between a and b in $>_i$. 

Theorem 3.7 (Arrow's Impossibility Theorem). *If $|A| \geq 3$, then the only (weak or strict) social welfare function that satisfy IIA and the Paretian property are dictatorial.*

3.2 Cardinal Aggregation

Assume each agent i has a utility function $u_i : A \rightarrow \mathbb{R}_+$. Assume $u_i \in U$. The most natural thing to do would be to provide axioms on a function

$$F : U^N \rightarrow \begin{cases} P \\ R \\ U \end{cases}.$$

Instead we will identify each $a \in A$ with the vector $(u_1(a), u_2(a), \dots, u_N(a)) \in \mathbb{R}_+^N$ and provide axioms on a binary relation $>_s$ on $\mathbb{R}_+^N$. In doing so we are assuming an outcome's desirability depends only on the utility profile.

Definition 3.8. A complete, transitive, and reflective binary relation $\succeq_s$ on $\mathbb{R}_+^N$ is called a **cardinal social welfare ranking**. 

Example 3.9.

- Utilitarian: $u \succeq u'$ if and only if $\sum_i u_i \geq \sum_i u'_i$.
- Rawlsian: $u \succeq u'$ if and only if $\min_i u_i \geq \min_i u'_i$. 

Definition 3.10. $\succeq_s$ is **continuous** if for any $u \in \mathbb{R}_+^N$ the sets $\{u' : u' \succeq_s u\}$ and $\{u' : u \succeq_s u'\}$ are closed. 

Equivalently, if $u^{(n)}$ and $v^{(n)}$ are such that $u^{(n)} \succeq_s v^{(n)}$, $u^{(n)} \rightarrow u$ and $v^{(n)} \rightarrow v$, then $u \succeq_s v$.

Definition 3.11. $\succeq_s$ is **independent of unconcerned agents** (IUA) if for any $x, y \in \mathbb{R}_+$ and any agent i , we have

$$(x, u_{-i}) \succeq_s (x, u'_{-i}) \iff (y, u_{-i}) \succeq_s (y, u'_{-i}), \quad \forall u_{-i}, u'_{-i}. \quad \text{img alt="book icon" data-bbox="760 565 780 580"/>$$

Theorem 3.12 (Debreu). *For any weak order $\succeq_s$ satisfying continuity and IUA, there exist functions $\{g_i\}_{i \in N}$ such that*

$$u \succeq_s u' \iff \sum_i g_i(u_i) \geq \sum_i g_i(u'_i).$$

Definition 3.13. $\succeq_s$ is **strictly monotonic** if whenever $u_i \geq u'_i$ for each i and there is some j with $u_j > u'_j$, then $u >_s u'$. 

Definition 3.14. $\succeq_s$ is **scale invariant** if $u \succeq_s u'$ if and only if $\lambda u \succeq_s \lambda u'$ for each $\lambda \in \mathbb{R}_{>0}$. 

Theorem 3.15. *If $\succeq_s$ satisfies continuity, IUA, strict monotonicity, and scale invariance, and if $N \geq 2$, then there are positive weights $w_1, \dots, w_N$ summing to 1 and some $\gamma \in \mathbb{R}$ such that*

$$u \succeq_s v \iff \sum_i w_i \frac{u_i^{1-\gamma} - 1}{1-\gamma} \geq \sum_i w_i \frac{v_i^{1-\gamma} - 1}{1-\gamma}.$$

Example 3.16 (A few limits).

- When $\gamma = 0$ we have weighted utilitarianism:

$$u \succeq_s v \iff \sum_i w_i u_i \geq \sum_i w_i v_i.$$

- When $\gamma \rightarrow 1$ we have $\lim(u^{1-\gamma} - 1)/(1 - \gamma) = \log u$, so

$$\begin{aligned} u \succeq_s v &\iff \sum_i w_i \log u_i \geq \sum_i w_i \log v_i \\ &\iff \prod_i u_i^{w_i} \geq \prod_i v_i^{w_i}. \end{aligned}$$

This is called **Nash welfare**.⁵

- When $\gamma \rightarrow \infty$, we get something called the **leximin order**:

$$u \succ_s v \iff u_k^* > v_k^*$$

where u^* is u rearranged into increasing order, similarly for v^* , and k is the smallest index where u and v differ. Thus this refines the Rawlsian order. 

3.3 Aggregation under Uncertainty

Suppose there is a finite number of states $\Theta = \{\theta_1, \dots, \theta_n\}$ and each agent i has a utility $u_i : \Theta \rightarrow \mathbb{R}$ and evaluates a lottery $p \in \Delta(\Theta)$ using the VNM expected utility $u_i(p) := \sum_k p_k u_i(\theta_k)$.

Remark 3.17. There are good normative reasons to use expected utility, but lab experiments show that people don't always do so. 

We assume the planner is also VNM rational with utility $u_s(p) := \sum_k p_k u_s(\theta_k)$. A weak requirement is that if $u_i(p) \geq u_i(q)$ for all i , then $u_s(p) \geq u_s(q)$. Call this a utility function satisfying the **Paretian property**.

Theorem 3.18 (Harsanyi). u_s is **Paretian** if and only if there is a constant β and non-negative weights $\{w_i\}_{i \in N}$ such that

$$u_s(\theta) = \beta + \sum_i w_i u_i(\theta), \quad \forall \theta \in \Theta.$$

This implies that

$$u_s(p) = \beta + \sum_i w_i u_i(p), \quad \forall p \in \Delta(\Theta).$$

To prove this, we will use the following version of Farkas's lemma:

Lemma 3.19. Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $B \in \mathbb{R}^{d \times n}$, and $c \in \mathbb{R}^d$. Then exactly one of the following holds:

- There exists $x \in \mathbb{R}^n$ such that $Ax \leq b$ and $Bx = c$.
- There exists $p \in \mathbb{R}^m$ and $q \in \mathbb{R}^d$, $p \geq 0$, such that $pA + qB = 0$ and $pb + qc < 0$.

⁵Often used in the CS literature. The outcome when you maximize Nash welfare is quite equal.

Proof (of Harsanyi Theorem). The conclusion is true if and only if there exists β and w_i such that

$$\begin{pmatrix} 1 & u_1(\theta_1) & \dots & u_N(\theta_1) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & u_1(\theta_n) & \dots & u_N(\theta_n) \end{pmatrix} \begin{pmatrix} \beta \\ w_1 \\ \vdots \\ w_N \end{pmatrix} = \begin{bmatrix} u_s(\theta_1) \\ \vdots \\ u_s(\theta_n) \end{bmatrix}$$

and

$$-\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix} \begin{pmatrix} \beta \\ w_1 \\ \vdots \\ w_N \end{pmatrix} \leq 0.$$

Now write A, B, c, p, q as in the statement of Farkas lemma above. If the conclusion does not hold, we can applying Farkas lemma to get $q_1, \dots, q_m$ and $p_1, \dots, p_N \geq 0$ such that

$$\sum_k q_k = 0, \quad \sum_k q_k u_j(\theta_k) = p_j, \quad \forall j, \quad \sum_k q_k u_s(\theta_k) < 0.$$

This is feasible if and only if the following is feasible:

$$\sum_k q_k = 0, \quad \sum_k q_k u_j(\theta_k) \geq 0, \quad \forall j, \quad \sum_k q_k u_s(\theta_k) < 0.$$

Note that q solves this system then λq solves it for each $\lambda > 0$ (the solution set of q is a cone). Thus scale q such that $|q_k| \leq 1/m$ for all k and define $r_k = q_k + 1/m$ for all k . Note that r_k is probability: $\sum_k r_k = 1$ and $r_k \geq 0$. The second and third equalities above imply

$$\begin{aligned} \sum_k r_k u_j(\theta_k) - \sum_k \frac{1}{m} u_j(\theta_k) &\geq 0, \quad \forall j \\ \sum_k r_k u_s(\theta_k) - \sum_k \frac{1}{m} u_s(\theta_k) &< 0. \end{aligned}$$

Thus everyone prefers r to the uniform lottery, but the social welfare ranking prefers the uniform lottery to r , contradicting the Paretian property. $\square$

4 Social Choice with Money

In many applications we may assume agents have quasi-linear utilities:

Definition 4.1. A **quasi-linear** utility on $A \times \mathbb{R}$ is one where the preference of (a, p) depends only on $u(a) - p$. 

Example 4.2. Consider the problem of allocating an indivisible object to one of two agents, each with a private value for the object. Assume each agent has quasi-linear utility.

Consider the following mechanism, called the **second price auction**: we ask each agent to tell us their value (privately), and the one with the higher value gets the object and pays us the second highest value. We claim that this mechanism is strategy-proof; that is, the dominant strategy for each agent is to report their true value.

To see this, suppose first that $v_1 < v_2$. There will be no change in the outcome if the deviation v'_1 is less than v_2 . If $v'_1 = v_2$, then player 1 wins with probability $1/2$ and gets utility $v_1 - v_2 < 0$ in the case of winning. If $v'_1 > v_2$, then player 1 wins with certainty but gets utility $v_1 - v_2 < 0$.

Next, suppose that $v_1 > v_2$ so that in the truth telling case player 1 wins and gets utility $v_1 - v_2 > 0$. If $v'_1 < v_2$, player 1 loses the good and get utility 0. If $v'_1 = v_2$ then player 1 wins with probability $1/2$ and gets utility $(v_1 - v_2)/2 < v_1 - v_2$. If $v'_1 > v_2$, the outcome does not change.

Finally, if $v_1 = v_2$, then any deviation will not change the resulting utility. 

Is this a good mechanism? Well, depends on the application. One potential downside is that you have to collect money and thus extract surplus from the agents.

Some justification for quasi-linear utilities:

Proposition 4.3. Suppose $\succeq$ is a weak preference on $A \times \mathbb{R}$, where A is a second-countable topological space.

Suppose $\succeq$ satisfies the following axioms:

- $(a, p) \succeq (a', p')$ if and only if $(a, p + \delta) \succeq (a', p' + \delta)$.
- For any $a, a' \in A$ there is a p such that $(a, p) \succeq (a', 0)$.
- For any a, p and $\delta > 0$ we have $(a, p) \succeq (a, p + \delta)$.⁶
- $\succeq$ is continuous.

Then there exists a function $u : A \times \mathbb{R} \rightarrow \mathbb{R}$ such that $(a, p) \succeq (a', p')$ if and only if $u(a) - p \geq u(a') - p'$. This function is said to **represent** $\succeq$.

We will next treat transfers as a goal to implement certain social choice function (this may be justified if we care only about total welfare, which is preserved under transfers, and not distributions). Formally, we want to **implement** some social choice function $g : U^N \rightarrow A$, where $U := \{u : A \rightarrow \mathbb{R}\}$ is the set of utility functions over A , using **transfers** $p : U^N \rightarrow \mathbb{R}^N$ such that if utilities are $u = (u_1, \dots, u_N)$, i pays $p_i(u)$ to the planner.

⁶Read p as the amount you need to pay.

4.1 Dominant Strategy Implementation

Throughout this section, we will be concerned with dominant strategy implementation:

Definition 4.4. We say g is **implementable in dominant strategies** if there exists p such that for each i, u_i, u_{-i}, u'_i , we have

$$u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) \geq u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}). \quad \text{📖}$$

4.2 Cyclical Monotonicity

Note that the constraints above can be rewritten as

$$p_i(u_i, u_{-i}) - p_i(u'_i, u_{-i}) \leq u_i(g(u_i, u_{-i})) - u_i(g(u'_i, u_{-i})).$$

Let's first fix a finite set $V \subset U$ so we can apply Farkas's lemma.

Consider the constraints for each i separately. Fix i and $v_{-i} \in V^{N \setminus \{i\}}$. Write $x(v) = p_i(v, v_{-i})$. The obedience constraints for i are

$$x(v) - x(v') \leq v(g(v, v_{-i})) - v(g(v', v_{-i})), \quad \forall v, v' \in V.$$

In matrix form this is $Ax \leq b$, where A has $|V| \times (|V| - 1)$ rows and $|V|$ columns with

$$A_{(v, v'), v''} = \begin{cases} 1 & \text{if } v'' = v \\ -1 & \text{if } v'' = v' \\ 0 & \text{otherwise} \end{cases}.$$

By Farkas's lemma, no transfers exist if and only if there exists $y \geq 0$ such that $yA = 0$ and $yb < 0$. We index y by pairs (v, v') . Note that $y \neq 0$ since otherwise $yb = 0$. There thus exists (v, v') such that $y_{(v, v')} > 0$, which implies the existence of v'' such that $y_{(v', v'')} > 0$ (otherwise the product of y and the column of A corresponding to v' would be negative). Repeating this argument, we get a sequence $v^1, v^2, \dots$ such that $y_{(v^k, v^{k+1})} > 0$ for each k . Since V is finite, the sequence eventually repeats an index. Let q be the first time it does, and assume without loss that $v^1 = v^q$. Now we have $y_{v^1, v^2} > 0, y_{v^2, v^3} > 0, \dots, y_{v^{q-1}, v^q} > 0, y_{v^q, v^1} > 0$. Let y^* be a zero-one vector such that $y^*_{(v, v')} = 1$ if and only if $(v, v') = (v^k, v^{k+1})$ for some k .

Note that $A_{(v^k, v^{k+1}), v^k} = 1$ which is cancelled by $A_{(v^{k-1}, v^k), v^k} = -1$ for each k in the collapsed yA . Thus by construction $y^*A = 0$. Let $\alpha := \min_k \{y_{(v^k, v^{k+1})}\}$. Note that $y - \alpha y^* \geq 0$ and $(y - \alpha y^*)A = 0$, but $y - \alpha y^*$ has at least one more 0 than y . Iterating, we arrive at the decomposition

$$y = \alpha_1 y_1^* + \alpha_2 y_2^* + \dots + \alpha_p y_p^*,$$

for some p , where each y_j^* is a **simple cycle** just like y^* . Since $yb < 0$, it must be the case that $y_\ell^* b < 0$ for some ℓ . Recall that y_ℓ^* is some zero-one vector. We have thus

shown that no transfers exists to implement g implies a sequence of utilities $w^1, \dots, w^r$ such that

$$\sum_{j=1}^{r-1} b_{(w^j, w^{j+1})} = \sum_{j=1}^{r-1} w^j(g(w^j, u_{-i})) - w^j(g(w^{j+1}, u_{-i})) < 0.$$

That is, a cycle of deviations is profitable (think two agents each wanting to be each other).

One can also show quite easily that the converse is also true: if the display above is violated, then there is a profitable deviation cycle, which is profitably for at least one of the agents. Thus we have:

Proposition 4.5 (Cyclical Monotonicity). *$g : V^N \rightarrow A$ is implementable in dominant strategies if and only if for each i , all v_{-i} , and each cycle $v^1, v^2, \dots, v^{q-1}, v^q = v^1$ we have*

$$\sum v^k(g(v^k, v_{-i})) - v^k(g(v^{k+1}, v_{-i})) \geq 0.$$

Remark 4.6. The negation of the condition above can be read as saying that there is a cycle of deviations (1 pretending to be 2, who pretends to be 3, ..., who pretends to be 1) that is on average profitable. ☕

It turns out this remains true if we replace V with U :

Theorem 4.7 (Rochet). *The same holds if $V = U$.*

4.3 Weak Monotonicity

Definition 4.8. Let $V \subset U$. We say $g : V^N \rightarrow A$ satisfies **weak monotonicity** if it satisfies cyclic monotonicity for all cycles of length 2. That is, for each i, v_{-i} and all v_i, v'_i , we have

$$v_i(g(v_i, v_{-i})) - v_i(g(v'_i, v_{-i})) + v'_i(g(v'_i, v_{-i})) - v'_i(g(v_i, v_{-i})) \geq 0.$$



This is clearly a necessary condition for implementability. It turns out that for most interesting applications it is also sufficient:

Theorem 4.9 (Saks and Yu, (2005)). *Suppose $V \subset U$ is convex. Then $g : V^N \rightarrow A$ is implementable in dominant strategies if and only if it satisfies weak monotonicity.*

In this sense, the cyclical monotonicity condition is not sensitive to domain restriction.

4.4 Welfare Maximization

As an application of the previous results, consider a welfare maximizing social choice function $g \in U^N \rightarrow A$:

$$g(u_1, \dots, u_N) \in \arg \min_{a \in A} \sum_{i \in N} u_i(a).$$

For any profile (u_i, u_{-i}) ,

$$u_i(g(u_i, u_{-i})) + \sum_{j \neq i} u_j(g(u_i, u_{-i})) \geq u_i(a) + \sum_{j \neq i} u_j(a), \quad \forall a \in A.$$

Thus,

$$u_i(g(u_i, u_{-i})) - u_i(a) \geq \sum_{j \neq i} u_j(a) - u_j(g(u_i, u_{-i})), \quad \forall a \in A.$$

Now fix u_{-i} and let $u^1, u^2, \dots, u^q = u^1$ be a cycle for i . We have

$$\begin{aligned} \sum_{\ell=1}^q u^\ell(g(u^\ell, u_{-i})) - u^\ell(g(u^{\ell+1}, u_{-i})) &\geq \sum_{\ell=1}^{q-1} \sum_{j \neq i} u_j(g(u^{\ell+1}, u_{-i})) - u_j(g(u^\ell, u_{-i})) \\ &\geq \sum_{j \neq i} \underbrace{\sum_{\ell=1}^{q-1} u_j(g(u^{\ell+1}, u_{-i})) - u_j(g(u^\ell, u_{-i}))}_{=0} = 0. \end{aligned}$$

Thus,

Corollary 4.10. *Any welfare maximizing social choice function is implementable in dominant strategies.*

Note that while the existence of transfers is established, we haven't said anything about how to find them. This is what we turn to next.

4.5 VCG Mechanisms

We are still in the setting of implementing a welfare maximizing social choice function $g : U^N \rightarrow A$. Consider the following payment rule:

$$p_i(u_i, u_{-i}) := - \sum_{j \neq i} u_j(g(u_i, u_{-i})),$$

so that each i gets compensated the total utility of the other agents. When reporting truthfully, i gets utility

$$\begin{aligned} u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) &= \sum_j u_j(g(u_i, u_{-i})) \\ &\geq \sum_j u_j(g(u'_i, u_{-i})) = u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}), \end{aligned}$$

where the inequality follows from the definition of g as a welfare maximizing social choice function. Note that this works because effectively every agent's utility function is the total welfare, which is perfectly aligned with the social welfare function.

This is expensive: you have to pay $N - 1$ times the total welfare! But note that the agents' incentives will not change under an additional transfer that is independent of their report. In particular, we have the following

Proposition 4.11. *Let $h : U^{N-1} \rightarrow \mathbb{R}$ be a function and p a payment rule implementing g . Then $p'_i(u_i, u_{-i}) = p_i(u_i, u_{-i}) + h(u_{-i})$, $p'_j \equiv p_j$, $\forall j \neq i$ also implements g .*

We may thus consider the following payment rule: For each i and each u_{-i} , let $a_i^*(u_{-i}) \in \arg \max_{a \in A} \sum_{j \neq i} u_j(a)$ denote the welfare maximizing action without i . Then, set p_i to be i 's **welfare externality**:

$$p_i(u_i, u_{-i}) := \sum_{j \neq i} u_j(a_i^*(u_{-i})) - u_j(g(u_i, u_{-i})).$$

As a direct corollary, we know that p_i implements g . One can see this directly as well:

$$\begin{aligned} u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) &= \sum_j u_j(g(u_i, u_{-i})) - \sum_{j \neq i} u_j(a_i^*(u_{-i})) \\ &\geq \sum_j u_j(g(u'_i, u_{-i})) - \sum_{j \neq i} u_j(a_i^*(u_{-i})) = u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}). \end{aligned}$$

Definition 4.12. The **Vickrey-Clarke-Groves (VCG)** mechanism has $g(u_1, \dots, u_N) \in \arg \max_{a \in A} \sum u_i(a)$ and

$$p_i(u_i, u_{-i}) = \sum_{j \neq i} u_j(a_i^*(u_{-i})) - u_j(g(u_i, u_{-i})).$$



The VCG mechanism is designed for U^N . What if we have some $V \subset U$ and a social choice function $g' : V^N \rightarrow A$? Note that the VCG mechanism restricted to V still implement g' in dominant strategies: if agents don't want to deviate in U , of course they don't want to deviate in V .

Example 4.13 (Second price auction). There is a single good to allocate to one of the agents. We identify A with the unit vectors in $\mathbb{R}^N$. Each agent i has a value b_i for the good and gets utility 0 if they don't get the good and b_i if they do (this is a domain restriction of the utilities). In the VCG mechanism, the agent with the highest value $b^1 := \max b_i$ gets the good, paying the second highest value b^2 . 

Example 4.14 (Sponsored search auctions). Suppose Google has k sponsored slots $1, \dots, k$. Each slot ℓ has a click through rate α_ℓ . Assume $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_k$. If $k < N$, add a bunch of "slots" with click through rate 0. Each firm $i = 1, \dots, N$ has value v_i from a click, so they get utility $\alpha_\ell v_i$ from slot ℓ (we have the domain restriction that utilities depend only on what you are assigned and not what others get). Assume $v_1 \geq v_2 \geq \dots \geq v_N$.

It turns out that the welfare maximizing assignment is to assign slot 1 to firm 1, ..., slot k to firm k . One can then easily verify that the VCG payments are $p_i(v_i, v_{-i}) = \sum_{j > i} (\alpha_{j-1} - \alpha_j) v_j$. 

The limits of dominant strategy implementation:

Theorem 4.15 (Robert). Suppose $|A| \geq 3$. Then $g : U^N \rightarrow A$ is implementable in dominant strategies if and only if there are nonnegative weights w_i , $i \in N$ not all zero, and constants c_a , $a \in A$ such that

$$g(u_1, \dots, u_N) \in \arg \max_{a \in A} \left[\left(\sum_{i \in N} w_i u_i(a) \right) + c_a \right], \quad \forall u_1, \dots, u_N.$$

Read c_a as biases for outcome a .

Remark 4.16. This result depends crucially on having unrestricted domain. 

5 Social Choice with Indifferences

Consider the following problems:

Problem 5.1 (Social choice with indifferences). For each $i \in N$, there is a partition $A^i = \{A_1^i, \dots, A_{N_i}^i\}$ of A such that i is indifferent between any two outcomes in the same cell of the partition. Let P^i be the set of strict preferences on A^i . Note that any strict preference on A^i induces a weak ranking on A . A social choice with indifferences problem is to find a social choice function

$$f : \prod_i P_i \rightarrow A.$$

This problem is defined by the tuple $(N, A, \{A^i\}_{i \in N})$. ✎

Problem 5.2 (Constrained allocation problem). N agents, each with a set of outcomes X_i . An **allocation** is a function $\mu : N \rightarrow \cup_i X_i$ such that $\mu(i) \in X_i$ for each i .

Let $\mathcal{A}$ be the set of all allocations. We impose constraints on the set of allocations represented by a nonempty set $C \subset \mathcal{A}$. Let $\tilde{P}_i$ be the set of strict preferences on X_i . A mechanism is a function $f : \prod_i \tilde{P}_i \rightarrow C$. The constrained allocation problem is defined by the tuple $(N, \{X_i\}_{i \in N}, C)$. ✎

It turns out that these two problems are equivalent:

Proposition 5.3. *For each social choice with indifferences problem, there is an equivalent constrained allocation problem and vice versa.*

Proof. Given $(N, (X_i)_i, C)$. We construct the equivalent social choice with indifferences problem as follows: Let $A := C$ and $A^i := \{\{\mu \in C : \mu(i) = x\} : x \in X_i\}$. A preference $>_i$ over X_i induces the preference $>_i^*$ over A^i in the obvious way, and any mechanism $f : \prod \tilde{P}_i \rightarrow C$ corresponds to the social choice function $f^* : \prod P_i \rightarrow A$ defined by $f^*(>_1, \dots, >_N) = f(>_1^*, \dots, >_N^*)$.

For the converse, fix $(N, A, \{A^i\}_i)$. Define $X_i := A_i$ and let C be the set of all $\mu : N \rightarrow \cup_i X_i = \cup_i A_i$ such that $\cap_i \mu(i) \neq \emptyset$. Any preferences over A_i carries over to X_i in the natural way. Any mechanism $f : \prod P_i \rightarrow A$ can be translated to the mechanism $f^* : \prod \tilde{P}_i \rightarrow C$ by setting $f^*(>)(i) = A_i^i$, where A_i^i contains $f(>)$. □

These two problems are mathematically equivalent. We will mainly work with the constrained allocation interpretation because that is how Root thought of this problem.

Example 5.4.

- House allocations: $X_i = H$ for each i .

$$C := \{\mu : N \rightarrow H \mid \mu(i) \neq \mu(j), \forall i \neq j\}.$$

- School choice: schools $X_i = X$ for each i . Each $x \in X$ has a capacity $q_x \geq 0$.

$$C := \{\mu : N \rightarrow X \mid |\mu^{-1}(x)| \leq q_x, \forall x\}.$$

- One-sided matching: $X_i = N$ for each i .

$$C := \{\mu : N \rightarrow N \mid \mu \circ \mu = \text{Id}\}.$$

- Two-sided matching: $N = M \cup W$, where M and W are disjoint. Set $X_i = W$ if $i \in M$ and $X_i = M$ if $i \in W$.

$$C := \{\mu : N \rightarrow N \mid \mu \circ \mu(i) = i, \forall \mu(i) \in M \text{ if } i \in W\}.$$

- Social choice: $X_i = A$ for each i .

$$C := \{\mu : N \rightarrow A \mid \mu(i) = \mu(j), \forall i, j\}.$$



Definition 5.5. A mechanism $f : \prod P_i \rightarrow C$ is **strategy-proof** if there exists no $i, >_i, >'_i, >_{-i}$ such that

$$f(>'_i, >_{-i})(i) >_i f(>_i, >_{-i})(i).$$

It is **Pareto efficient** if for each $> \in \prod P_i$, there exists no $\mu \in C$ such that

$$\mu(j) \succeq_j f(>)(j), \forall j,$$

with the inequality above strict for some j .



We now present a characterization of strategy-proof and Pareto efficient mechanisms for the constrained allocation problem, first in the two agent case, then in the general case.

5.1 The Two Agent Case

Consider the two agent case.

Proposition 5.6. Let f be Pareto efficient and strategy-proof. If for some $x \in X_1$ there is no $\mu \in C$ such that $\mu(i) = x$, then i 's ranking of x does not affect the outcome.

Proof. Suppose $>_i, >'_i$ differ only in the position of x . We claim that by strategy-proofness, $f(>_i, >_j)(i) = f(>'_i, >_j)(i)$. To see this, note that if $f(>_i, >_j)(i) >_i f(>'_i, >_j)(i)$, then since neither outcome is x , we have the same ranking under $>'_i$, contradicting strategy-proofness. The other case is immediate.

This implies that $f(>_i, >_j)(j) = f(>'_i, >_j)(j)$, for if not, one of these allocations is inefficient. $\square$

Thus we may restrict attention to the case where for each $x \in X_i$ there exists $\mu \in C$ such that $\mu(i) = x$. In particular, writing each allocation as a pair (x, y) , we know that for each infeasible (x, y) , there exists x' and y' such that $(x', y) \in C$ and $(x, y') \in C$.

Suppose (x, y) is not feasible and x', y' are as above. Suppose further that preferences are as follows:

$$\begin{aligned} x &>_1 x' >_1 \dots \\ y &>_2 y' >_2 \dots \end{aligned}$$

Note that this setup is similar to that in the Gibbard-Satterthwaite theorem, and we will similarly use the technique of propagating the ambiguity in the outcome under

this preference profile. By Pareto efficiency, either $f(>_1, >_2) = (x, y')$ or $f(>_1, >_2) = (x', y)$. Suppose it is the former and take any preference $>'_2$ that top ranks y . Then $f(>_1, >'_2)(2) \neq y$ by strategy-proofness, and so $f(>_1, >'_2)(1) = x$ by efficiency (otherwise this outcome will be Pareto dominated by (x', y)). Thus, $f(>'_1, >'_2)(i) = x$ for each $(>'_1, >'_2)$ such that 1 top ranks x and 2 top ranks y . We say 1 is the **local dictator** at (x, y) .

The same argument shows that there is a local dictator, say $D(x, y)$, at each infeasible $(x, y) \in \bar{C} := \mathcal{A} \setminus C$.

Proposition 5.7. $D(x, y) = 1$ implies $D(x, y') = 1$ for each y' such that $(x, y') \in \bar{C}$ and $D(x', y) = 1$ for each x' such that $(x', y) \in \bar{C}$.

Proof. We prove the first part; the second is symmetric.

Suppose not, so that $D(x, y') = 2$. Then at the preference profile

$$\begin{aligned} x &>_1 \dots \\ y &>_2 y' >_2 \dots, \end{aligned}$$

agent 2 gets something worse than y and y' , since 1 is the local dictator at (x, y) and neither (x, y) nor (x, y') is feasible. But 2 will simply deviate to $y' >'_2 \dots$, getting $f(>_1, >'_2)(2) = y'$. $\square$

Let G be the graph with one node for each infeasible allocation and an edge between any two infeasible allocations which share a coordinate. As a corollary of the proposition above, each connected component of G has a local dictator. From this we easily recover Gibbard-Satterthwaite: either the top ranked outcome is feasible (each agent gets what she wants), or the top ranked outcome is infeasible (the single local dictator gets what she wants).

6 Auctions

Example 6.1.

- (i) Art, used cars, etc.
- (ii) Electricity.
- (iii) Google/Facebook.
- (iv) Electromagnetic spectrum.
- (v) Government debt.



Example 6.2 (Warmup). Single buyer for a single good with value $v \in [0, \bar{v}]$. Suppose v has differentiable cdf F with density f . Suppose for simplicity that the seller has no value for the good.



In this example, for a posted price p , the expected revenue is $p(1 - F(p))$. The FOC for maximizing revenue gives

$$p = \frac{1 - F(p)}{f(p)}.$$

Note that the lowest type that's willing to buy at the optimal price p is exactly p , and at that type, $v - (1 - F(v))/f(v) = 0$.

We now consider a more general setup:

- N agents.
- Single good.
- Each agent has value $v \in [0, 1]$.
- A set of outcomes $\{(x_1, \dots, x_N) : x_i \geq 0, \forall i, \sum_i x_i \leq 1\}$ describing the probability of each agent getting the good, with some probability of the good not being allocated.
- Agent i gets utility $vx_i - p_i$ if they get the good with probability x_i and pays p_i .

Think of this as two domain restrictions of the general model: (1) quasilinear utilities, (2) all kinds of indifferences, including the fact that each agent cares only about their own allocation probability.

Example 6.3 (Auctions).

- (i) First price auction: the seller asks each agent i to submit a bid $b_i \geq 0$. The player with the highest bid wins and pays their bid.
- (ii) Second price auction: the player with the highest bid wins and pays the second highest bid.
- (iii) All-pay auction: player with highest bid wins and everyone pays their bid.



6.1 Revenue in First and Second Price Auctions

We will suppose $N = \{1, 2\}$ and $v_i \stackrel{\text{iid}}{\sim} F$, where F admits density f which is strictly positive on $[0, 1]$.

6.1.1 Second Price Auction

Recall that we've seen that bidding your own value is a dominant strategy. Let us thus assume that $b_i = v_i$ for each i .

Note that the probability of the lower value being less than or equal to some w is $1 - (1 - F(w))(1 - F(w)) = 2F(w) - F(w)^2 =: F_2(w)$. Denote the derivate of this function as $f_2(w)$. Then the expected revenue is

$$\int_0^1 x f_2(x) dx = w F_2(w) \Big|_0^1 - \int_0^1 F_2(w) dw = 1 - \int_0^1 F_2(w) dw.$$

6.1.2 First Price Auction

Remark 6.4 (A Review of Bayesian Games). N agents. Agent i has a set of types T_i and a set of actions A_i . Write $T := \prod T_i$ and $A := \prod A_i$. Agent i has a utility function

$$u_i : T_i \times A \rightarrow \mathbb{R}.$$

Assume a common prior $\mu \in \Delta(T)$. A **strategy** for i is a function $\sigma_i : T_i \rightarrow \Delta(A_i)$. Let $\sigma = (\sigma_i)_{i \in N}$ denote a strategy profile. Note that

$$u_i(\sigma, t_i) = \sum_{t_{-i}, a} u_i(t_i, a) \sigma_i(a_i | t_i) \sigma_{-i}(a_{-i} | t_{-i}) \mu(t_{-i} | t_i).$$

We say a strategy profile σ is a **Bayes Nash equilibrium** if

$$u_i(\sigma, t_i) \geq u_i(\sigma'_i, \sigma_{-i}, t_i), \quad \forall i, t_i, \sigma, \sigma'_i.$$



Let's guess that there is an equilibrium of the first price auction where there is a function $b : [0, 1] \rightarrow [0, 1]$ such that, in equilibrium, if either bidder has value v , they bid $b(v)$. Assume that b is strictly increasing and differentiable.

A necessary condition for b to be a BNE is: if i has value v_i , then their utility $(v_i - b(v'_i))F(v'_i)$ must be maximized at $v'_i = v_i$. The FOC gives

$$-b'(v'_i)F(v'_i) + (v_i - b(v'_i))f(v'_i) \Big|_{v'_i=v_i} = 0,$$

or

$$\underbrace{b'(v_i)F(v_i) + b(v_i)f(v_i)}_{\frac{d}{dv_i} b(v_i)F(v_i)} = v_i f(v_i).$$

Integrating both sides gives

$$b(v)F(v) = \int_0^v t f(w) dw = vF(v) \Big|_0^v - \int_0^v F(w) dw.$$

Thus,

$$b(v) = v - \frac{1}{F(v)} \int_0^v F(w) dw.$$

Note that the distribution of the higher value is $F_1(w) = F(w)^2$. Let $f_1(w)$ denote the derivative of F_1 . The revenue in a first price auction is thus

$$\begin{aligned}\int_0^1 b(v)f_1(v) dv &= b(v)F_1(v)\Big|_0^1 - \int_0^1 b'(v)F_1(v) dv \\&= b(1) - \int_0^1 \frac{f(v)}{F(v)^2} \int_0^v F(w) dw F_1(v) dv \\&= 1 - \int_0^1 F(w) dw - \int_0^1 f(v) \int_0^v F(w) dw dv \\&= 1 - \int_0^1 F(w) dw - \left[F(v) \int_0^v F(w) dw \right]_0^1 + \int_0^1 F(v)^2 dv \\&= 1 - 2 \int_0^1 F(w) dw + \int_0^1 F(w)^2 dw = 1 - \int_0^1 F_2(w) dw,\end{aligned}$$

which is the same as the revenue in the second price auction.

7 Revenue Maximizing Auctions

Suppose the seller designs a game G to maximize revenue, which specifies the following:

- For each $i \in N$, an action set $\mathcal{A}_i$. Write $\mathcal{A} := \prod_j \mathcal{A}_j$.
- An allocation function $x : \mathcal{A} \rightarrow A$ and a payment rule $p : \mathcal{A} \rightarrow \mathbb{R}^N$.

Suppose that each $i \in N$ comes up with a strategy $s_i : [0, 1] \rightarrow \Delta \mathcal{A}_i$. (We still assume each agent's value $v_i \sim F$ on $[0, 1]$.) Assume $(s_i)_i$ is a BNE of the game G .

Fix $(s_i)_i$. Let's focus on one agent i and drop subscripts. Let $X(v)$ and $P(v)$ respectively denote the probability that i gets the good when they have value v and the expected payment they make when they have value v . These are called the **interim** allocation and payment rule. Note that i has expected utility

$$vX(v) - P(v).$$

Since s is a BNE, we have for arbitrary v, v' that

$$\begin{aligned} vX(v) - P(v) &\geq vX(v') - P(v') \\ v'X(v) - P(v) &\geq v'X(v') - P(v'). \end{aligned}$$

This gives the **payment difference sandwich**:

$$v'(X(v) - X(v')) \leq P(v) - P(v') \leq v(X(v) - X(v')).$$

Thus, $(v - v')(X(v) - X(v')) \geq 0$, so X is nondecreasing. But note also the graphical interpretation of the payment difference sandwich above, which motivates the following:

Let $0 = v_0 < v_1 < \dots < v_k < v_{k+1} = v$. Note that

$$P(v) - P(0) = P(v) - P(v_k) + P(v_k) - P(v_{k-1}) + \dots + P(v_1) - P(0).$$

Thus, applying the payment difference sandwich repeatedly gives

$$\sum_{j=0}^k v_j (X(v_{j+1}) - X(v_j)) \leq P(v) - P(0) \leq \sum_{j=0}^k v_{j+1} (X(v_{j+1}) - X(v_j)).$$

Note that both bounds are like the Riemann integral of X^{-1} . We will avoid the potential problem of X not having an inverse by viewing the integral of X^{-1} as the whole box subtracted off the integral of X .

Note that

$$\begin{aligned} \sum_{j=0}^k v_j (X(v_{j+1}) - X(v_j)) &= vX(v) - vX(v) + \sum_{j=0}^k v_j (X(v_{j+1}) - X(v_j)) \\ &= vX(v) - v_{k+1}X(v_{k+1}) + \sum_{j=0}^k v_j X(v_{j+1}) - \sum_{j=0}^{k-1} v_{j+1} X(v_{j+1}) \\ &= vX(v) + \sum_{j=0}^k v_j X(v_{j+1}) - \sum_{j=0}^k v_{j+1} X(v_{j+1}) \\ &= vX(v) - \sum_{j=0}^k (v_j - v_{j+1}) X(v_{j+1}). \end{aligned}$$

Similarly telescoping the other bound, we can conclude that

$$vX(v) - \sum_{j=0}^k (v_{j+1} - v_j)X(v_{j+1}) \leq P(v) - P(0) \leq vX(v) - \sum_{j=0}^k (v_{j+1} - v_j)X(v_j).$$

Now both sides are actual Riemann sums. Taking limits on both sides,

$$vX(v) - \int_0^v X(t) dt \leq P(v) - P(0) \leq vX(v) - \int_0^v X(t) dt.$$

Thus, from the payment difference sandwich, we can conclude the following:

Proposition 7.1.

- (i) $X(v)$ is nondecreasing.
- (ii)

$$P(v) - P(0) = vX(v) - \int_0^v X(t) dt.$$

Definition 7.2. An auction game G with equilibrium s is **individually rational** if for each v , $vX(v) - P(v) \geq 0$. 

Thus in particular, we must have $P(0) \leq 0$. We set $P(0) = 0$ to maximize revenue. Then,

$$P(v) = vX(v) - \int_0^v X(t) dt$$

depends only X . Thus we have

Proposition 7.3 (Revenue Equivalence). *For any G, G' and equilibria s, s' such that X_G and $X_{G'}$ are the same, revenue is the same.*

7.1 Revelation Principle

Definition 7.4. A direct **mechanism** is a game where $\mathcal{A}_i = V$, so that agents report a value v . 

Proposition 7.5 (Revelation Principle). *For any game G and equilibrium s , there is a direct mechanism where it is a BNE for agents to truthfully report such that the outcome is the same under G and s as under the direct mechanism and truthful reporting.*

Proof. Write

$$G = ((\mathcal{A}_i)_i, x, p), \quad s = (s_i)_i, \quad M = ((V)_i, x', p'),$$

where we define the direct mechanism M as follows:

$$\begin{aligned} x'(v_1, \dots, v_N) &:= x(s_1(v_1), \dots, s_N(v_N)) \\ p'(v_1, \dots, v_N) &:= p(s_1(v_1), \dots, s_N(v_N)). \end{aligned}$$

□

7.2 Optimal Auctions

Thus it is without loss to use direct mechanisms with $\mathcal{A}_i = [0, 1]$, and where we choose

$$x : [0, 1]^N \rightarrow A, \quad p : [0, 1]^N \rightarrow \mathbb{R}^N$$

Let f^k be the density of k draws from F . The interim allocation and payment are

$$\begin{aligned} X_i(v_i) &:= \int_{v_{-i}} x_i(v_i, v_{-i}) f^{N-1}(v_{-i}) dv_{-i} \\ P_i(v_i) &:= \int_{v_{-i}} p_i(v_i, v_{-i}) f^{N-1}(v_{-i}) dv_{-i}. \end{aligned}$$

We know that $X_i(\cdot)$ must be nondecreasing. The seller's problem is

$$\max \int_v \left(\sum_i p_i(v_i, v_{-i}) \right) f^N(v_i, v_{-i}) dv,$$

subject to the constraints that X_i is nondecreasing and $P_i(v_i) = v_i X_i(v_i) - \int_0^{v_i} X_i(t) dt$. We can rewrite the problem as

$$\begin{aligned} \max \sum_i \int_v p_i(v_i, v_{-i}) f^N(v_i, v_{-i}) dv &= \sum_i \int_0^1 P_i(v_i) f(v_i) dv_i \\ &= \sum_i \int_0^1 \left(v_i X_i(v_i) - \int_0^{v_i} X_i(t) dt \right) f(v_i) dv_i, \end{aligned}$$

where the second integral can be rewritten as

$$\begin{aligned} \int_0^1 f(v_i) \left(\int_0^{v_i} X_i(t) dt \right) dv_i &= F(z) \int_0^z X_i(t) dt \Big|_0^1 - \int_0^1 F(v_i) X_i(v_i) dv_i \\ &= \int_0^1 X_i(t) dt - \int_0^1 F(v_i) X_i(v_i) dv_i \\ &= \int_0^1 (1 - F(v_i)) X_i(v_i) dv_i. \end{aligned}$$

Thus, the revenue becomes

$$\sum_i \int_0^1 \underbrace{\left(v_i - \frac{1 - F(v_i)}{f(v_i)} \right)}_{\text{virtual value}} X_i(v_i) f(v_i) dv_i = \int_v \sum_i \left(v_i - \frac{1 - F(v_i)}{f(v_i)} \right) x_i(v) f^N(v) dv.$$

Write the **virtual value** as $c(v_i)$. Thus the auctioneer wants to set $x_i(v_i, v_{-i}) = 1$ for the highest value v_i , and 0 otherwise, as long as the virtual value is positive:

$$x_i^*(v) := \begin{cases} 0, & c(v_i) \leq 0 \text{ or if } c(v_i) < c(v_j) \text{ for some } j \neq i \\ \frac{1}{m}, & \text{otherwise, where } m := |\{j : v_j = \max c(v_j)\}| \end{cases}.$$

This candidate allocation rule is implementable if each interim allocation function $X_i^*(v_i)$ is nondecreasing. This is not always the case, since the virtual value is not always nondecreasing.

Definition 7.6. A distribution F is **regular** if $w - (1 - F(w))/f(w)$ is strictly increasing. 

Note that F being regular implies that the candidate x^* is nondecreasing in each component, and thus each X_i^* is nondecreasing and thus implementable. We now need to choose p^* such that

$$P_i^*(v_i) = v_i X_i^*(v_i) - \int_0^{v_i} X_i^*(t) dt.$$

The easiest thing to do is of course to set

$$p_i^*(v_i, v_{-i}) := v_i x_i^*(v_i, v_{-i}) - \int_0^{v_i} x_i^*(t, v_{-i}) dt.$$

We have now fully specified a candidate optimal mechanism.

Let's characterize this mechanism. Let $\underline{v}$ be the unique (recall that c is strictly increasing) value such that $c(\underline{v}) = 0$ if one exists, other set $\underline{v} := 0$. We have the following cases:

- (i) $v_i < \max\{\underline{v}, v_j, \forall j \neq i\} =: s$. Then $x_i(v_i, v_{-i}) = 0$ and you pay

$$p_i(v_i, v_{-i}) = v_i \cdot 0 - \int_0^{v_i} \underbrace{x_i(t, v_{-i})}_0 dt = 0.$$

- (ii) $v_i > s$. Then $x_i(v_i, v_{-i}) = 1$ and you pay

$$p_i(v_i, v_{-i}) = v_i \cdot 1 - \int_0^{v_i} x_i(t, v_{-i}) dt = v_i - \int_s^1 1 dt = v_i - s.$$

- (iii) $v_i = s \neq \underline{v}$, and suppose m people have the highest value. Then $x_i(v_i, v_{-i}) = 1/m$ and you pay

$$p_i(v_i, v_{-i}) = v_i \cdot \frac{1}{m} - \int_0^{v_i} x_i(t, v_{-i}) dt = \frac{v_i}{m}.$$

This is precisely the second price auction with reserve price $\underline{v}$.

Some “side-benefits”:

- (i) We asked for BNE and got dominant strategies.
- (ii) We asked for interim IR and got ex post IR.
- (iii) If $N = 1$, we get posted price at $\underline{v}$.

7.3 The Multi-Good Case

Now, consider the two goods case and suppose there is only a single buyer. Assume that the buyer has private values v^1 and v^2 respectively for each of the two goods, so that if they get probability x^1 of good 1 and probability x^2 of good 2 by paying p , then their utility is

$$v^1 x^1 + v^2 x^2 - p.$$

In writing so, we are assuming additive valuations and quasilinear utilities. Assume that $v^1 \sim F^1$ and $v^2 \sim F^2$ independently.

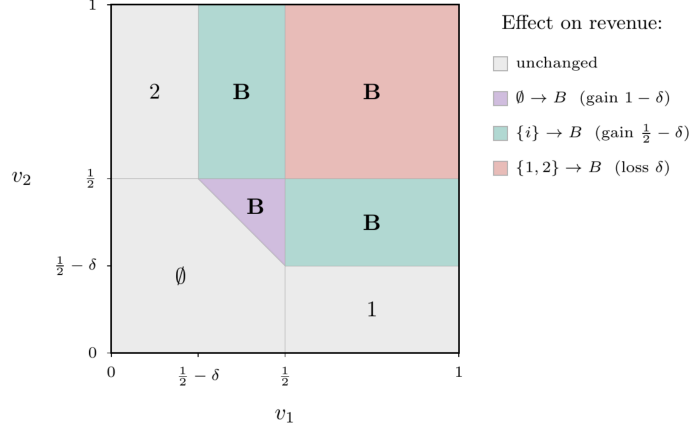


Figure 1: The Effect of Introducing a Bundle B at Price $p = 1 - \delta$ on Revenue.

Remark 7.7. One might think that this problem decouples into two single-good problems, but this is not the case. To see this consider the one bidder, two good case with $v^i \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1]$. Optimal posted price for each good separately is $1/2$. Suppose the seller introduces the bundle of both goods at a price $p = 1 - \delta$. The buyer will buy the bundle if and only if

- The value of the bundle is at least the price $v_1 + v_2 > p$.
- It's better to buy the bundle than to buy good 1 alone: $v_1 + v_2 - p > v_1 - 1/2$.
- It's better to buy the bundle than to buy good 2 alone: $v_1 + v_2 - p > v_2 - 1/2$.

The latter two inequalities simplify to $v_2 > p - 1/2$ and $v_1 > p - 1/2$ respectively. We plot the effect of the introduction of this new bundle in Figure 1. Thus the introduction of the bundle changes by

$$\frac{\delta^2}{2}(1 - \delta) + \delta \left(\frac{1}{2} - \delta \right) - \frac{\delta}{4}.$$

Note that with $\delta = 0$, the bundle has the same price as the two separate goods, so the bundle doesn't change anything. With small δ , the quadratic and cubic terms are smaller than the linear term, and we get to first order an additional revenue of $\delta/4 > 0$. ☕

It turns out that we can find the optimal auction under the following strong assumption:

Assumption 7.8. F^1 and F^2 are uniform on $[0, 1]$.

In this case, the optimal mechanism is the following:

- The bidder can either buy either good at price $3/4$, or
- buy both goods at price $(4 - \sqrt{2})/3$.

For any other distributions, we know next to nothing about the optimal mechanism, but we do know that they can be quite complicated.

More generally, a mechanism can offer a menu of the form

$$\{z_\alpha^1, z_\alpha^2, z_\alpha^{12}, p_\alpha\}_\alpha.$$

If F^1 and F^2 are Beta distributions, the optimal mechanism offers an uncountably large menu.

7.4 The Mussa-Rosen Model

Setup:

- Single buyer and seller.
- Seller sells a single good but can choose the quality $q \in [0, \underline{q}]$ by paying a cost $c(q)$ for producing quality q . Assume $c(0) = 0$ and $c'(q), c''(q) > 0$ for each q , so that c is strictly increasing and strictly convex.
- The buyer has a value $\theta \in [0, \bar{\theta}]$ and is drawn from a distribution with cdf F and density f .
- If the buyer has type θ and buys a good of quality q at the price p , her utility is $\theta q - p$, and the seller gets $p - c(q)$.

Let's appeal to the revelation principle. The seller gets to choose a direct mechanism, described by $q : [0, \bar{\theta}] \rightarrow \mathbb{R}$ and $p : [0, \bar{\theta}] \rightarrow \mathbb{R}$. We can repeat the analysis before to conclude that q must be nondecreasing and

$$p(\theta) = \theta q(\theta) - \int_0^\theta q(t) dt.$$

Thus for a fixed direct mechanism, we can write the expected revenue as

$$\int_0^{\bar{\theta}} \left(\theta q(\theta) - \int_0^\theta q(t) dt \right) f(\theta) d\theta.$$

The second (nested) integral can be rewritten as

$$\begin{aligned} \int_0^{\bar{\theta}} f(\theta) \int_0^\theta q(t) dt d\theta &= F(\bar{z}) \int_0^{\bar{z}} q(t) dt \Big|_0^{\bar{\theta}} - \int_0^{\bar{\theta}} F(\theta) q(\theta) d\theta \\ &= \int_0^{\bar{\theta}} (1 - F(\theta)) q(\theta) d\theta. \end{aligned}$$

Thus the expected profit is

$$\int_0^{\bar{\theta}} \left[\left(\theta - \frac{1 - F(\theta)}{f(\theta)} \right) q(\theta) - c(q(\theta)) \right] f(\theta) d\theta.$$

As before, we choose q to maximize the integrand for each θ . The derivative of the integrand with respect to q is

$$\theta - \frac{1 - F(\theta)}{f(\theta)} - c'(q).$$

We assume F is regular and let $\underline{\theta}$ be the largest value such that

$$\theta - \frac{1 - F(\theta)}{f(\theta)} - c'(\theta) \leq 0.$$

We set $q^*(0) = 0$ if $\theta < \underline{\theta}$. Otherwise, since c' is strictly increasing, for each $\theta > \underline{\theta}$ there exists a unique $q^*(\theta)$ such that

$$\theta - \frac{1 - F(\theta)}{f(\theta)} - c'(q^*(\theta)) = 0.$$

Let's contrast this with what a benevolent planner would do. The planner knows θ and chooses q to maximize the sum of the utilities $q\theta - c(q)$. The FOC gives $\theta - c'(q^*) = 0$. This is not in general the same as the optimal q^* above: in general q^* is smaller than the planner's choice. But as $\theta \rightarrow \bar{\theta}$, the virtual value approaches $\bar{\theta}$ and thus $q^*(\theta)$ approaches the planner's choice.

Proposition 7.9. $q^* \leq q^*$, and $q^*(\theta) \rightarrow q^*(\theta)$ as $\theta \rightarrow \bar{\theta}$.

This result is sometimes called “no distortion at the top.”

8 Cooperative game with Transferable Utility

Example 8.1. Three cities A, B, C , each needs to build a public project (e.g., airport, water treatment plant, etc.). They could cooperate or do it alone, with the following costs:

- A alone: $c(A) = 120$.
- B alone: $c(B) = 140$.
- C alone: $c(C) = 120$.
- $c(AB) = 170$.
- $c(AC) = 160$.
- $c(BC) = 190$.
- $c(ABC) = 255$.



One can verify that

$$c(ABC) < c(A) + c(B) + c(C)$$

$$c(ABC) < c(A) + c(BC)$$

$$c(ABC) < c(B) + c(AC)$$

$$c(ABC) < c(C) + c(AB).$$

So the efficient outcome is for A, B, C to all cooperate. But how much should each pay? Each pay $c(ABC) = 85$? But then A and C are collectively paying more than $c(AC)$.

What about each pay their stand-alone cost less the average savings $(380 - 255)/3$? But then A and B will be paying more than $c(AB)$.

Let x_A, x_B, x_C be the amount paid by A, B, C respectively. To sustain cooperation, we have the following constraints:

$$x_A \leq c(A) = 120$$

$$x_B \leq c(B) = 140$$

$$x_C \leq c(C) = 120$$

$$x_A + x_B \leq c(AB) = 170$$

$$x_A + x_C \leq c(AC) = 160$$

$$x_B + x_C \leq c(BC) = 190$$

$$x_A + x_B + x_C = c(ABC) = 255.$$

These are called the **no subsidy constraints**.

We will switch variables to simplify this system. Write the savings as

$$y_A := c(A) - x_A, \quad y_B := c(B) - x_B, \quad y_C := c(C) - x_C.$$

Then the system above is equivalent to

$$\begin{aligned} y_A, y_B, y_C &\geq 0 \\ y_A + y_B &= c(A) + c(B) - (x_A + x_B) \geq 90 \\ y_B + y_C &\geq 70 \\ y_A + y_C &\geq 80 \\ y_A + y_B + y_C &= 125. \end{aligned}$$

A solution is $y_A = 55, y_B = 35, y_C = 35$.

Now suppose we change $c(ABC)$ to 265. Then the new system is the same, with only the last constraint now being changed to $y_A + y_B + y_C = 115$. But note that adding the two-term constraints gives

$$2[y_A + y_B + y_C] \geq 240 > 2 \cdot 115,$$

so there is no solution. There is no way to sustain cooperation without screwing over at least one person.

Definition 8.2. A **cooperative game with transferable utility** is a pair (N, v) , where N is a set of agents, and $v : 2^N \setminus \{\emptyset\} \rightarrow \mathbb{R}$ is a function that assigns a value to each nonempty coalition of agents.

An **allocation** is a vector $x \in \mathbb{R}^N$ such that $\sum_{i \in N} x_i = v(N)$. The **core** of (N, v) is the set of allocations

$$\left\{ x \in \mathbb{R}^N : \forall \emptyset \neq S \subset N, \sum_{i \in S} x_i \geq v(S) \right\}. \quad \text{📖}$$

Example 8.3. $N = \{1, 2, 3\}$. We can write the inequalities defining the core as

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \geq \begin{pmatrix} v(1) \\ v(2) \\ v(3) \\ v(12) \\ v(13) \\ v(23) \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = v(123).$$



In the general case, the core is defined by $Ax \geq v, 1 \cdot x = v(N)$

$$A \in \{0, 1\}^{(2^N - 2) \times N}, \quad A_{Sj} = \begin{cases} 1, & j \in S \\ 0, & \text{otherwise} \end{cases}$$

A version of Farkas's lemma gives the following: The core is nonempty if and only if there is no $p \geq 0, q \in \mathbb{R}$ such that $p(-A) + q1^\top = 0$ and $p(-v) + qv(N) < 0$.

It is easy to see that q can only be strictly positive. Thus we can divide through, writing $\delta = p/q \geq 0$, to get

$$\delta A = 1^\top, \quad \delta v > v(N).$$

Here, δ is called a **balanced** set of weights, i.e., a function $\delta : 2^N \setminus \{\emptyset, N\} \rightarrow [0, 1]$ such that for each i , $\sum_{S: i \in S} \delta(S) = 1$. Thus each agent is in coalitions with total weight 1. The core is nonempty if and only if there is some weights δ such that $\delta v > v(N)$. In the latter case, the artificial coalition have too much value.

Example 8.4. In the motivating example, we have $\delta(AB) = \delta(AC) = \delta(BC) = 1/2$. 

This result can be summarized as follows:

Theorem 8.5 (Bondareva-Shapley). *A cooperative game with transferable utility (N, v) has a nonempty core if and only if*

$$\sum_{\emptyset \neq S \subseteq N} \delta(S) v(S) \leq v(N)$$

for every balanced set of weights δ .

9 Cooperative Games with Nontransferable Utility

Definition 9.1. A cooperative game with nontransferable utility (NTU) is specified by a set N of agents, and for each coalition $S \subset N$, a set A_S of “actions” that can be taken by S . Each $i \in N$ has a weak ranking over $\bigcup_{S:i \in S} A_S$.

We say $a \in A_N$ is in the **core** if there is no coalition $S \subset N$ and $a' \in A_S$ such that $a' \succeq_j a$ for each $j \in S$ and $a' \succ_i a$ for some $i \in S$. 

There aren't a lot of things we can say about general NTU. Instead, we will examine a few specific problems in NTU.

9.1 One-Sided Matching

The one-sided matching consists of N agents who we want to match up into pairs and singletons. Each $i \in N$ has a strict ranking on N , $\succ_i$, and we interpret $i \succ_i j$ to mean i prefers to be unmatched than to be matched with j . A **matching** is a function $\mu : N \rightarrow N$ such that $\mu \circ \mu = \text{Id}$. For each coalition, we set

$$A_S := \{\nu : S \rightarrow S \mid \nu \circ \nu = \text{Id}_S\}.$$

We can extend each $\succ_i$ on N to a ranking on $\bigcup_{S:i \in S} A_S$ by letting

$$\nu \succeq_i \eta \iff \begin{cases} \nu(i) \succ_i \eta(i), \text{ or} \\ \nu(i) = \eta(i) \end{cases}.$$

Then, a matching μ is in the core if no coalition S can deviate to a matching ν that makes everyone in S at least as well off and someone in S strictly better off.

Example 9.2 (The Core can be Empty). Consider the following preferences:

$$\begin{aligned} 2 &\succ_1 3 \succ_1 1 \\ 3 &\succ_2 1 \succ_2 2 \\ 1 &\succ_3 2 \succ_3 3, \end{aligned}$$

and everyone prefers to be matched with someone than to be in a singleton. It is easy to verify that the core is empty. (By symmetry suffices to consider the case where 1 is matched with 2.) 

Definition 9.3.

- If $j \succ_i i$, we say j is **acceptable** to i .
- A matching μ is **individually rational** if $\mu(i)$ is acceptable for every i such that $\mu(i) \neq i$.
- A matching μ is **stable** if it's individually rational and there are no i, j such that $j \succ_i \mu(i)$ and $i \succ_j \mu(j)$. Such a pair (i, j) is called a **blocking pair**.



In this case, only coalitions of size 2 matter:

Proposition 9.4. *A matching μ is in the core if and only if it is stable.*

Proof. $\Rightarrow$: easy.

$\Leftarrow$: Suppose μ is not in the core, so there exists a coalition $S \subset N$ and $\nu \in A_S$ such that

$$\nu(i) \succeq_i \mu(i), \quad \forall i \in S$$

with strict inequality for some $i \in S$. We have the following cases:

- $\nu(i) = i$. Then $i \succ \mu(i)$. Thus μ is not individually rational, and thus not stable.
- $\nu(i) = k \neq i$. Note that k is in S so $\nu(k) \succeq_k \mu(k)$. But since k 's matching changed from $\mu(k) \neq i$ (unequal else i 's matching never changed, which is impossible) to $\nu(k) = i$, we have $\nu(k) \succ_k \mu(k)$, so i, k is a blocking pair. $\square$

9.2 Two-Sided Matching

Suppose M, W are nonempty sets that partition N . We can view this as a special case of the one-sided matching problem by letting each $m \neq m' \in M$ have

$$m \succ_m m', \quad \forall m' \in M \setminus \{m\},$$

and likewise for each $w \in W$, so that it is unacceptable for an agent to be matched with someone on the same side.

It turns out that the core is always nonempty in this case, and accordingly there is a stable matching.

Definition 9.5. A *m-proposing deferred acceptance* (DA) algorithm proceeds as follows:

- Step 1: All $m \in M$ propose to their favorite acceptable w . Each $w \in W$ rejects all but their favorite *acceptable* proposer (if any), to whom they are tentatively matched.
- Step k : All $m \in M$ rejected in the last round propose to their favorite acceptable w who has not yet rejected them. Each $w \in W$ considers their tentative match from the previous step and any new proposers. They reject any unacceptable and all but the favorite acceptable, to whom they are tentatively matched.

The algorithm terminates when there is no more proposal, when everyone is matched to their tentative matches. 

We have the following immediate observations:

Proposition 9.6.

- *The welfare of each w is nondecreasing during the course of the algorithm.*
- *The algorithm runs in at most $|M| \cdot |W| + 1$ steps.*

Example 9.7. $M = \{m_1, m_2, m_3, m_4\}$ and $W = \{w_1, w_2, w_3, w_4\}$. We will list their preferences only for acceptable matches:

$\succ_{m_1}$	$\succ_{m_2}$	$\succ_{m_3}$	$\succ_{m_4}$	$\succ_{w_1}$	$\succ_{w_2}$	$\succ_{w_3}$	$\succ_{w_4}$
w_1	w_1	w_1	w_1	m_1	m_1	m_3	m_1
	w_2	w_2	w_2		m_4		m_2
		w_3	w_3		m_2		
			w_4		m_3		

Round 1: All m proposes to w_1 . m_2, m_2, m_3 rejected. Tentative match: (m_1, w_1) .
 Round 2: m_2, m_3, m_4 propose to w_2 . m_2, m_3 rejected. Tentative matches: $(m_1, w_1), (m_4, w_2)$.
 Round 3: m_3 proposes to w_3 . No rejection. Tentative matches: $(m_1, w_1), (m_4, w_2), (m_3, w_3)$.



Proposition 9.8. *The m -proposing DA outcome is stable.*

Proof. Let μ_{DA} denote the resulting matching from DA. Note that μ_{DA} is IR since no agents are ever matched with an unacceptable partner at any point in the algorithm. We next check stability. Take any w, m not matched under μ_{DA} and suppose $w \succ_m \mu_{DA}(m)$. Then, m proposed to w before he was matched with $\mu_{DA}(m)$. Since $\mu_{DA}(w) \neq m$, w must have rejected m , so it is impossible that $m \succ_w \mu_{DA}(w)$. $\square$

It turns out that the m -proposing DA outcome is the best stable matching for all m and the worst stable matching for all w .

Definition 9.9. Say w is **achievable** for m if there exists a stable matching where m and w are matched. 

Proposition 9.10. *In the m -proposing DA outcome, each $m \in M$ is matched with their most preferred achievable w (except perhaps for those who are unmatched). Each w is matched with their least preferred achievable m (except perhaps for those who are unmatched).*

Proof. We first show that each m is matched with their most preferred achievable w . Suppose not. Consider the first round of the DA algorithm where an $m \in M$ was rejected by an achievable w . Since (m, w) are matched in some stable matching ν , they are mutually acceptable. From w rejecting m in DA, we know that she is at this point tentatively matched to some m' , who by assumption has not been rejected by any achievable partner. Thus, any achievable partner of m' is ranked below w . In particular, we have $w \succ_{m'} \nu(m')$, so ν is not stable, a contradiction.

It remains to show that there does not exist a w and an achievable m such that $m' := \mu(w) \succ_w m$. Suppose this m, w pair exists, so that $\nu(m) = w$ for some stable matching ν . Note that $\nu(m') \neq w$ so by what has just been proven, $\mu(m') = w \succ_{m'} \nu(m')$. Thus (m', w) is a blocking pair for ν , a contradiction. $\square$

9.3 Mechanism Design for Two-Sided Matching

We now specialize the mechanism design problem to the two-sided matching problem.

For each $m \in M$, let P_m be the set of strict preferences on $W \cup \{m\}$, and define P_w symmetrically. Let P be the set of preference profiles where $\succ_m \in P_m$ for each m and $\succ_w \in P_w$ for each w . Let Σ be the set of matchings:

$$\Sigma := \left\{ \mu : M \cup W \rightarrow M \cup W \left| \begin{array}{l} \mu(m) \in W \cup \{m\}, \forall m \\ \mu(w) \in M \cup \{w\}, \forall w \end{array} \right. , \mu \circ \mu = \text{Id} \right\}.$$

Definition 9.11. A **mechanism** is a function $f : P \rightarrow \Sigma$. We say a mechanism is **stable** if for every $\succ \in P$, $f(\succ)$ is stable. 

Proposition 9.12. *There are no stable and strategy-proof mechanisms.*

Proof. Consider the following preferences:

$\succ_{m_1}$	$\succ_{m_2}$	$\succ_{w_1}$	$\succ_{w_2}$
w_1	w_2	m_2	m_1
w_2	w_1	m_1	m_2

The only stable matchings are $A := \{(m_1, w_1), (m_2, w_2)\}$ and $B := \{(m_1, w_2), (m_2, w_1)\}$ (these are coincidentally what we get by running the m - and w -preferred DA). Let f be a stable matching. Then $f(\succ) \in \{A, B\}$. Assume WLOG $f(\succ) = A$. Then m_2 can deviate by listing m_1 as unacceptable, so that the only stable matching at the deviated profile is B . $\square$

However, the m -proposing DA is always stable and strategy-proof for all m . This is much more difficult to prove, but let's plow through it.

We will prove this result under the simplifying assumption that all m, w are mutually acceptable and $|M| = |W|$. The result extends to the general case with some more work.

Lemma 9.13 (Blocking Lemma). *Let μ_{DA} be the outcome of the m -proposing DA mechanism with preferences $\succ$. Let μ be any other matching and write $M' := \{m \in M \mid \mu(m) \succ_m \mu_{\text{DA}}(m)\}$. Then*

- (i) $M' \neq M$, so there are no matching that can make all m strictly better off.
- (ii) If $M' \neq \emptyset$, then there is a $m \in M \setminus M'$ and a $w \in \mu(M')$ such that (m, w) is a blocking pair for μ .

Note that for each stable μ , $M' = \emptyset$ by the DA property.

Proof. We will consider the following two cases:

Case 1: $\mu(M') \neq \mu_{\text{DA}}(M')$. Then $M' \neq M$ is immediate, otherwise $\mu(M') = \mu(M) = W = \mu_{\text{DA}}(M) = \mu_{\text{DA}}(M')$. Note that the same argument shows $M' \neq \emptyset$.

For (ii), let $w \in \mu(M') \setminus \mu_{\text{DA}}(M')$. Let $m := \mu_{\text{DA}}(w)$. We will show that (w, m) is a blocking pair. By assumption, $m \notin M'$ (otherwise $w \in \mu_{\text{DA}}(M')$), so by definition of M' we have $\mu(m) \prec_m w$. The preference here is strict because the matching for m changed from w to someone else, as w is currently matched with someone in M' .

Thus it remains to show that $m \succ_w \mu(w)$. This must be true since otherwise, $\mu(w) \succ_w m = \mu_{\text{DA}}(w)$ and combined with the fact that $w \succ_{\mu(w)} \mu_{\text{DA}}(\mu(w))$ (from $\mu(w) \in M'$ and definition of M'), we know that $(w, \mu(w))$ blocks μ_{DA} , which contradicts the stability of μ_{DA} .

Case 2: $\mu(M') = \mu_{\text{DA}}(M')$. That is, the m 's who are made better off are just swapping partners among themselves.

Let k be the first step of the m -proposing DA where all agents in M' are matched with their partners from μ_{DA} . Let m' be an arbitrary agent in M' matched at this step, say with $w := \mu_{\text{DA}}(m')$. Write $m'' := \mu(w)$. Given the case we are considering, $m'' \in M'$,

so $w \succ_{m''} \mu_{\text{DA}}(m'')$. Thus m'' proposed to w at some point and was rejected by her at some step weakly before step k .

Now let m be the agent to whom w was matched at the beginning of step k . Since m was rejected in step k (with w being matched instead to m'), he is matched in a later step, so $m \notin M'$. Note also that $m \succ_w m'' = \mu(w)$, since m'' was rejected first.

The pair (m, w) is our proposed blocking pair. To establish this, it remains to show $w \succ_m \mu(m)$.

First note that since m proposed to w in the DA algorithm and was rejected, we have $w \succ_m \mu_{\text{DA}}(m)$. Furthermore, since $m \notin M'$, we have $\mu_{\text{DA}}(m) \succeq_m \mu(m)$. In either case, we have $w \succ_m \mu(m)$.

Note that in proving part (ii), part (i) is also proved, since we found $m \notin M'$. $\square$

Theorem 9.14. *The m -proposing DA mechanism is stable and strategy-proof for agents in M .*

Proof. Pick any $\succ$ and a deviation $\succ'_m$ for m . Let μ and ν respectively be the outcomes of the DA mechanism under $\succ$ and $(\succ'_m, \succ_{-m})$. Suppose for contradiction that $\nu(m) \succ_m \mu(m)$. By the blocking lemma, there must be a blocking pair (m', w') of ν with $m' \neq m$ (m being in M'), under the preference $\succ$. Since m' and w' don't change preference from $\succ$ to $(\succ'_m, \succ_{-m})$, we see that (m', w') also blocks ν under $(\succ'_m, \succ_{-m})$. This contradicts the fact that ν is stable (as the DA outcome) under $(\succ'_m, \succ_{-m})$. $\square$

9.4 Many-to-One Matching

We extend the analysis of one-to-one matching to many-to-one matching. Now, each w has a capacity $c_w \geq 1$, which we think of as the **capacity** of w . Suppose each $w \in W$ has a strict preference on $\{M' \subset M : |M'| \leq c_w\}$.

In this setting, a **matching** is a function $\mu : M \rightarrow M \cup W$ such that $|\mu^{-1}(w)| \leq c_w$ for each w , and $\mu(m) = m'$ implies $m = m'$.

We say a matching μ is **stable** if no m is matched to an unacceptable w and there is no set $M' \subset M$ such that

$$M' \succ_w \mu^{-1}(w)$$

and each $m \in M'$ weakly prefers w to $\mu(m)$. The pair (M', w) is called a **blocking pair**. That is, w cannot find a set of agents that she prefers to her current match, with each agent finding w at least as good as their current match.

Note that this condition implies IR for W : in a stable matching, no w is matched with an unacceptable set of agents.

A stable matching in this case may not exist.

Example 9.15. $M = \{m_1, m_2\}$, $W = \{w_1, w_2\}$, $c_{w_1} = 2, c_{w_2} = 1$. Suppose preferences are as follows:

$\succ_{m_1}$	$\succ_{m_2}$	$\succ_{w_1}$	$\succ_{w_2}$
w_2	w_1	$\{m_1, m_2\}$	$\{m_2\}$
w_1	w_2	$\{m_1\}$	$\{m_1\}$
m_1	m_2	$\emptyset$	$\emptyset$

None of the matchings are stable:

$\mu(m_1)$	$\mu(m_2)$	Blocked by
m_1	m_2	$\{m_2\}, w_2$
m_1	w_2	$\{m_1\}, w_1$
w_1	w_2	$\{m_1, m_2\}, w_1$
w_1	w_1	$\{m_1\}, w_2$
w_2	w_1	$\emptyset, w_1$
w_2	m_2	$\{m_2\}, w_2$



The culprit here is the complementary of w_1 's preferences. We shall rule such cases out:

Definition 9.16. A preferences $\succ_w$ is **responsive**⁷ if there is a strict ranking $\succ_w^*$ on $M \cup \{w\}$ such that for any S with $|S| < c_w$,

- for any $m, m' \notin S$, $S \cup \{m\} \succ_w S \cup \{m'\}$ if and only if $m \succ_w^* m'$;
- $S \cup \{m\} \succ_w S$ if and only if $m \succ_w^* w$.



With responsive preferences, stable matching exists and is reached with a natural extension of the DA algorithm:

Algorithm 9.17. First denote $\mu_0(m) = m$ for each m .

- Step 1. All m apply to their favorite acceptable w . If there are no applications, return μ_0 . Otherwise, each $w \in W$ considers the set of applicants A'_w and rejects all $m \notin \arg \max_{\succ_w} \{M' \subset A'_w : |M'| \leq c_w\}$. That is, w picks her favourite subset of applicants. Let μ_1 be the resulting matching.
- Step $k \geq 2$. All m apply to their favorite acceptable w who has not yet rejected them. Each w gets applicants A_w^k and again rejects all $m \notin \arg \max_{\succ_w} \{M' \subset A_w^k : |M'| \leq c_w\}$. Let μ_k be the resulting matching. If there is no rejections in step k , return μ_k .

Exercise 9.18. Show DA returns a stable matching if each $\succ_w$ is responsive.

It turns out that responsiveness is much stronger than necessary. Even with the DA algorithm, conditions like **rejection monotonicity** (agents who are rejected will remain rejected under expansion of applications pool) or observable rejection monotonicity (previous condition for applicants in the DA algorithm) suffices to guarantee the existence of a stable matching.

9.5 Is DA a Good Mechanism?

To fix ideas, let's focus on the context of college admissions. Recall that we've shown that the student-proposing DA is stable and strategy-proof for students, and the best stable matching for students. We've also shown in the problem set that this matching is Pareto efficient for both parties. However, one thinks that the college's preferences should not matter, but the DA outcome can be quite bad for the students if we truly disregard the colleges' preferences. In particular, it can be Pareto inefficient for the students.

⁷Not a great name, but it stuck.

Example 9.19. Consider the following preferences:

$\succ_{m_1}$	$\succ_{m_2}$	$\succ_{m_3}$	$\succ_{m_n}$	$\succ_{m_{n+1}}$	$\succ_{w_1}$	$\succ_{w_2}$	$\succ_{w_n}$
w_1	w_2	w_3	w_n	w_n	m_2	m_3	m_1
w_2	w_3	w_4	w_1	w_1	m_3	m_4	m_2
w_3	w_4	w_5	w_2	w_2	m_4	m_5	m_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
w_{n-1}	w_n	w_1	w_{n-2}	w_{n-2}	m_n	m_{n+1}	m_{n-1}
w_n	w_1	w_2	w_{n-1}	w_{n-1}	m_{n+1}	m_1	m_n
					m_1	m_2	m_{n+1}

One can show that each m is matched with their least preferred w . Though the empirical literature seems to suggest that this kind of preferences is not common. 

One might thus consider mechanisms tailored to preferences with more structure. In practice, two overwhelming factors that students care about are (1) school quality, and (2) distance. The preferences are generally decreasing in distance and decreasing in test scores.

Definition 9.20. We say preferences $(\succ_m)_{m \in M}$ and $(\succ_w)_{w \in W}$ are **aligned** if there exists $u : M \times W \rightarrow \mathbb{R}$ such that $u(m, \bullet)$ represents m 's preferences and $u(\bullet, w)$ represents w 's preferences for each m, w . 

Example 9.21. Suppose u is given by

u	w_1	w_2	w_3
m_1	7	6	8
m_2	1	2	0
m_3	4	5	7

Then m_1 prefers w_3 to w_1 to w_2 , while w_1 prefers m_1 to m_3 to m_2 , and so on.

In this case the DA matching is (m_1, w_3) , (m_3, w_2) , (m_2, w_1) . Welfare maximizing matching is the diagonal. Leximin matching is also the diagonal. In particular, in the aligned case DA behaves like a greedy algorithm, which can be worse than the “optimal” matching. 

Now suppose each m has utility of the form $-d(m, w) + q(w)$, where d stands for distance and q is the quality. Suppose also that each w has preference $-d(m, w) + h(m)$. Then, the preferences are aligned with $u(m, w) = -d(m, w) + q(w) + h(m)$. This suggests that preferences in school choice is at least approximately aligned, so DA may be “too greedy.”

10 Allocation Problems

Setup:

- A set X of “objects” (think houses).
- N agents, each with a strict preference $\succ_i$ on X .
- Assume for simplicity $|N| = |X|$.
- An **allocation** is a bijection $\mu : N \rightarrow X$. Let A denote the set of allocations.
- A **mechanism** is a function $f : P^N \rightarrow A$.

Definition 10.1. We say a mechanism f is

- **strategy-proof** if

$$f(\succ)(i) \succeq_i f(\succ'_i, \succ_{-i})(i), \quad \forall \succ, i, \succ'_i.$$

- **group strategy-proof** if for all $\succ \in P$ there is no coalition $S \subset N$ and $\succ'_S = (\succ'_i)_{i \in S}$ such that

$$f(\succ'_S, \succ_{-S})(j) \succeq_j f(\succ)(j), \quad \forall j \in S$$

with the preference being strict for at least one $j \in S$.

- **monotonic** if for any $\succ, \succ'$ such that

$$f(\succ)(i) \succ_i x \implies f(\succ')(i) \succ'_i x, \quad \forall i, x,$$

we have $f(\succ') = f(\succ)$.

- **non-bossy** if for all i and $\succ_{-i}$ with $f(\succ_i, \succ_{-i})(i) = f(\succ'_i, \succ_{-i})(i)$, we have

$$f(\succ_i, \succ_{-i}) = f(\succ'_i, \succ_{-i}).$$



Lemma 10.2. For any mechanism the following are equivalent:

- f is group strategy-proof.
- f is strategy-proof and non-bossy.
- f is monotonic.

Proof. Exercise. □

Lemma 10.3. If a mechanism f is group strategy-proof, then it is Pareto-efficient if and only if it's surjective.

Proof. Exercise. □

Definition 10.4. The **serial dictatorship** mechanism with order $\rho : N \rightarrow \{1, \dots, |N|\} \rightarrow N$, with ρ a bijection, is the mechanism that gives $\rho(1)$ their top choice, $\rho(2)$ their favorite among the remaining objects, and so on. 📖

Next, we consider a condition called neutrality, which basically says the mechanism should not depend on the names of the objects.

Let σ be a permutation of X . If $\mu \in A$, define $\sigma\mu$ to be the allocation where each i gets $\sigma(\mu(i))$. If $\succ_i \in P$, let $\sigma \succ_i$ be the preference $\succ'_i$ such that $x \succ'_i y$ if and only if $\sigma(x) \succ_i \sigma(y)$.

Definition 10.5. A mechanism f is **neutral** if for each $\succ \in P^N$ and all σ , we have

$$f(\sigma \succ) = \sigma f(\succ).$$



Example 10.6. Consider the following preferences:

	$\succ_1$	$\succ_2$	$\succ_3$
	a	a	b
	b	c	a
	c	b	c
ρ	1	2	3

Then serial dictatorship with order ρ gives a to agent 1, c to agent 2, and b to agent 3. One can see that this mechanism is neutral. 

Theorem 10.7 (Svensson). *Serial dictatorships are the only group strategy-proof and neutral mechanisms.*

Proof. $\Rightarrow$ is straightforward. We show $\Leftarrow$.

Let f be group strategy-proof and neutral. Fix $\succ^* \in P$ and consider the profile $(\succ^*, \succ^*, \dots, \succ^*)$. Suppose $\succ^*$ ranks x_1 first x_2 second and so on. There exists i such that i gets x at this profile. Without loss, call this agent 1. Similarly, assume 2 gets x_2 , and so on. By neutrality, 1 gets their top choice at any profile where all agents agree. So on profiles where all n agents have the same ranking, f agrees with serial dictatorship with order $\rho = \text{Id}$.

Now, let $\succ$ be an arbitrary preference profile. Let μ be the serial dictatorship outcome under $\succ$ and let $\succ'$ be the preference profile where all agents top rank $\mu(1)$, second rank $\mu(2)$, etc. Note that $f(\succ') = \mu$. We will convert $\succ'$ to $\succ$ step by step and show that the outcome is preserved at each step.

Consider $f(\succ_1, \succ', \dots, \succ')$. We have $f(\succ_1, \succ', \dots, \succ')(i) = \mu(1)$ by strategy-proofness since $\succ_1$ and $\succ'$ both rank $\mu(1)$ first. By non-bossiness, we have $f(\succ_1, \succ', \dots, \succ') = f(\succ', \dots, \succ') = \mu$.

Now consider $f(\succ_1, \succ_2, \succ', \dots, \succ')$. By strategy-proofness, the outcome for 2 is weakly better than $\mu(2)$, which is what 2 gets in serial dictatorship. Thus under $\succ_2$, either 2 top-ranks $\mu(2)$ or they top-rank $\mu(1)$ and second-rank $\mu(2)$. It cannot be the case that

$$f(\succ_1, \succ_2, \succ', \dots, \succ')(2) = \mu(1),$$

since with preference $\succ'$, 2 gets $\mu(2)$, so 2 would deviate to $\succ'$. Thus we have $f(\succ_1, \succ_2, \succ', \dots, \succ')(2) = \mu(2)$, and by non-bossiness,

$$f(\succ_1, \succ_2, \succ', \dots, \succ') = \mu.$$

We finish the proof by induction. □

Let's consider another mechanism, called **top trading cycles** (TTC), which can be thought of as the following algorithm:

Step 0: Given $\succ \in P^N$. Start with an initial allocation μ_0 .

Step 1: Each agent points to the agent who is assigned your favorite object under μ_0 . Formally, this is a graph. Since we have one edge out of each node, there must be at least a cycle.⁸ We trade endowments along all cycles. Remove the agents and objects allocated in this step.

Step k : Repeat until stable.

Example 10.8. Consider the following preferences:

	$\succ_1$	$\succ_2$	$\succ_3$
	a	a	b
	b	c	a
	c	b	c
μ_0	b	c	a
μ_1	a	c	b

The first step graph is $1 \leftrightarrow 3 \leftarrow 2$. We will show that TTC is group strategy-proof and Pareto efficient, so by Svensson's theorem, it is not neutral.

Indeed, consider the permutation $\sigma(c) = a$, $\sigma(b) = b$, and $\sigma(a) = c$. Then, noting that μ_0 does not change, we see that there is no nontrivial cycle.

	$\succ_1$	$\succ_2$	$\succ_3$
	c	c	b
	b	a	c
	a	b	a
μ_0	b	c	a



Proposition 10.9. *TTC is group strategy-proof and Pareto efficient.*

Proof. It suffices to show TTC is monotonic and surjective. To see surjectivity, make each agent top rank their assigned object. Then, expand the TTC graph to make each agent point to their favorite object, and each object point to its owner under the initial assignment μ_0 . This is a bipartite graph, with each node having one edge in and one edge out (bijection of μ and μ_0), so it decomposes into cycles.

Next, we show monotonicity. Suppose $\text{TTC}_{\mu_0}(\succ) = \mu$ and let $\succ'$ be a profile such that for all i and all x , if $\mu(i) \succ_i x$, then $\mu(i) \succ'_i x$. We need to show $\text{TTC}_{\mu_0}(\succ') = \mu$. Let $N_1, N_2, \dots, N_q$ be a partition of N such that N_k is the set of agents who were assigned their object in round k of $\text{TTC}_{\mu_0}(\succ)$. We will prove by induction that each cycle in $\text{TTC}_{\mu_0}(\succ)$ also forms under $\text{TTC}_{\mu_0}(\succ')$ in a weakly earlier round.

First consider the base case $k = 1$. Since each agent in N_1 gets their top choice, which does not change from $\succ$ to $\succ'$, Thus the same cycle forms in the first step of $\text{TTC}_{\mu_0}(\succ')$.

Next, before considering the inductive step, observe that for any i choosing in round k , any x such that $x \succ'_i \mu(i)$ is assigned to an agent (and thus removed) in a round strictly

⁸Each tree has one fewer edges than nodes.

earlier than i in $\text{TTC}(>)$. This is because $x \succ'_i \mu(i)$ implies $x \succ_i \mu(i)$ by monotonicity and then $\mu_0^{-1}(x) \in N_\ell$ for some $\ell < k$ (else i would point to it in round k).

Now let's tackle the inductive step. Assume that for all $\ell < k$, all cycles in N_ℓ has formed in $\text{TTC}_{\mu_0}(>')$ weakly before round ℓ . Consider a cycle $C := j_1 \rightarrow j_2 \rightarrow \dots \rightarrow j_S \rightarrow j_1$ in N_k . For any j in this cycle — if she has yet to exit in $\text{TTC}_{\mu_0}(>')$ — and any $x \succ'_j \mu(j)$, we must have $x \succ_j \mu(j)$, so that $\mu_0^{-1}(x) \in N_\ell$ for some $\ell < k$ (else j would point to it in $\text{TTC}_{\mu_0}(>)$). By the induction hypothesis, all such x have been allocated to agents other than j under $\text{TTC}_{\mu_0}(>')$. Thus either each j_s in the cycle points to j_{s+1} , so that this cycle will form, or some j_s has exited. In the latter case, let $r < k$ be the first round in which any member of the cycle exits, and let D be the cycle in round r containing that member. In this round, each $\mu(j_s)$ is still available, so if $D \neq C$, some j_s points to $x \succ'_{j_s} \mu(j_s)$, which by monotonicity satisfies $x \succ_{j_s} \mu(j_s)$. Thus, $\mu_0^{-1}(x) \in N_\ell$ for some $\ell < k$ (else j_s would point to it in round k of $\text{TTC}_{\mu_0}(>)$), which by the induction hypothesis means that it exited with its original cycle in $\text{TTC}_{\mu_0}(>)$ in a round weakly before ℓ . This cannot be true since $j_s \in N_k$ is a part of this cycle. $\square$

10.1 Random Allocation

Serial dictatorship and TTC are kind of unfair. Let's try randomization.

We can describe any allocation $\mu \in A$ with a $N \times N$ permutation matrix $M(\mu)$ defined by

$$M(\mu)(i, a) := \begin{cases} 1, & \mu(i) = a \\ 0, & \text{otherwise} \end{cases}.$$

Going forward let's identify μ and $M(\mu)$.

Definition 10.10. A square matrix M is **bistochastic** if all entries are nonnegative and the rows and columns sum to 1. 

Let B_N denote the set of $N \times N$ bistochastic matrices. It is easy to see that permutation matrices are bistochastic, and that B_N is convex. It turns out that the extreme points of B_N are exactly the permutation matrices:

Theorem 10.11 (Birkhoff-von Neumann). *Any bistochastic matrix is a convex combination of permutation matrices.*

Definition 10.12. A (random) allocation mechanism is a function $f : P^N \rightarrow B_N$, where $f(>)_{ia}$ is i 's probability of being assigned object a when preferences are $>$. 

Write $f(>)_i$ for the i th row of the matrix $f(>)$. Note that by Birkhoff von Neumann, there exists a lottery over A which yields $f(>)$ for all $>$, called a **social implementation** of $f(>)$.

Remark 10.13. We could have defined random assignment mechanisms as $f : P^N \rightarrow \Delta(A)$. The way we defined it makes it clear that i 's preference depends only on $f(>)_i$. 

We will assume that agents' preferences are consistent with the first-order stochastic dominance relation.

Definition 10.14. Let $P, P' \in \Delta(X)$ and let $\succ_i$ rank the objects in the order $x_1, x_2, \dots$. We say P **first-order-stochastically dominates** P' with respect to $\succ_i$ if

- $P(x_1) \geq P'(x_1)$,
- $P(x_1) + P(x_2) \geq P'(x_1) + P'(x_2)$, and so on, so that
- $\sum_{y: y \succeq_i x} P(y) \geq \sum_{y: y \succeq_i x} P'(y)$ for all x .

In this case we write $P \text{FOSD}_{\succ_i} P'$. 

Thus if we shift weights from worse objects to better objects, we get a lottery that is strictly better in the sense of first-order stochastic dominance.

Proposition 10.15. $P \text{FOSD}_{\succ_i} P'$ if and only if $\mathbb{E} u(P) \geq \mathbb{E} u(P')$, for any u that agree with $\succ_i$.

Definition 10.16. We say f is **strategy-proof** if

$$f(\succ_i, \succ_{-i})_i \text{FOSD}_{\succ_i} f(\succ'_i, \succ_{-i})_i, \quad \forall i, \succ, \succ'_i.$$



Note that this is satisfied by lotteries over strategy-proof deterministic mechanisms, such as RSD and RTTC.

Definition 10.17. We say f is **ex post efficient** if for all $\succ$, $f(\succ)$ has a social implementation $\alpha_1 \mu_1 + \dots + \alpha_p \mu_p$ where each μ_j is Pareto efficient. 

Definition 10.18. Say f satisfies **equal treatment of equals** (ETOE) if for any i, j with $\succ_i = \succ_j$, we have

$$f(\succ_i, \succ_j, \succ_{-ij})_i = f(\succ_i, \succ_j, \succ_{-ij})_j, \quad \forall \succ_{-ij}.$$



Definition 10.19. A **random serial dictatorship** is the mechanism

$$\text{RSD}(\succ) := \frac{1}{N!} \sum_{\rho} \text{SD}_{\rho}(\succ),$$

where SD_{ρ} is the serial dictatorship mechanism with order ρ . 

Exercise 10.20. RSD is strategy-proof, ex post efficient, satisfies ETOE, and neutral.

Definition 10.21. A **random top trading cycle** is the mechanism

$$\text{RTTC}(\succ) := \frac{1}{N!} \sum_{\mu_0} \text{TTC}_{\mu_0}(\succ),$$

where TTC_{μ_0} is the TTC mechanism with initial allocation μ_0 . 

Proposition 10.22. $\text{RSD} = \text{RTTC}$.

Is RSD the unique strategy-proof, ex post efficient, and ETOE mechanism? One can prove this for $N = 3$ by hand and $N = 4$ by Claude, but it turns out that this is false for $N = 5$.

Example 10.23.

$$\begin{array}{c|c|c|c}
 \succ_1 & \succ_2 & \succ_3 & \succ_4 \\
 \hline
 a & a & b & b \\
 b & b & a & a \\
 c & c & c & c \\
 d & d & d & d
 \end{array} , \quad \text{RSD}(\succ) = \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{pmatrix} a & b & c & d \\ \frac{5}{12} & \frac{1}{12} & \frac{1}{4} & \frac{1}{4} \\ \frac{5}{12} & \frac{1}{12} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{12} & \frac{5}{12} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{12} & \frac{5}{12} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

The last two columns follows from ETOE. For the first two columns, consider the case where 1 gets a . This happens if

- 1 is first dictator (with probability $1/4$),
- 1 is the second dictator and 3 or 4 is the first dictator (with probability $(1/2) \cdot (1/3) = 1/6$).

Recall also that a previous result shows that this is ex post efficient, but the following allocation gives a FOSD improvement for each agent:

$$\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Thus ex post efficiency does not imply efficiency in the ordinary sense. 

Ordinary efficiency is captured by the notion of ordinal efficiency.

Definition 10.24. We say f is **ordinally efficient** if for every $\succ$ there is no $M \neq f(\succ)$ such that

$$M_i \text{ FOSD}_{\succ_i} f(\succ)_i, \quad \forall i. \quad \text{📖}$$

Definition 10.25. The **probabilistic serial mechanism** PS works as follows: Let time t move from 0 to 1. At each instant of time t , each agent “eats” dt probability share of their favorite house that is not yet exhausted. 

Example 10.26.

$$\begin{array}{c|c|c|c}
 \succ_1 & \succ_2 & \succ_3 & \succ_4 \\
 \hline
 a & a & b & b \\
 b & b & a & a \\
 c & c & c & c \\
 d & d & d & d
 \end{array} , \quad \text{PS}(\succ) = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Between time $t = 0$ and $t = 1/2$, 1 and 2 eat a , and 3 and 4 eat b . At $t = 1/2$, objects a and b are exhausted. Between $t = 1/2$ and $t = 3/4$, all agents eat c . At time $t = 3/4$, c is exhausted, and so all agents eat d for the remaining time. 

Exercise 10.27. PS is ordinally efficient.

Example 10.28.

$$\begin{array}{c|cccc}
 & \succ_1 & \succ_2 & \succ_3 & \succ'_3 \\
 \hline
 a & a & b & a \\
 b & c & a & b \\
 c & b & c & c
 \end{array}, \quad \text{PS}(\succ) = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{2} & 0 & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{2} \end{pmatrix}, \quad \text{PS}(\succ') = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{6} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}.$$

Now, $\text{PS}(\succ)_3$ and $\text{PS}(\succ')_3$ is not comparable in the FOSD sense, so PS is not strategy-proof. It turns out, however, that a first order improvement is not possible. 

It is easy to see that PS satisfies ETOE. One might wonder if ETOE, ordinal efficiency, and strategy-proofness are compatible. The answer is negative.

Theorem 10.29 (Bogomolnaia and Moulin, 2001). *There is no strategy-proof, ordinally efficient, and ETOE mechanism.*

11 Information Design

Example 11.1. Two agents: a prosecutor (sender) and a judge (receiver). There is a defendant who both the prosecutor and the judge believe to be guilty with probability 0.3. Prosecutor gets utility 1 if the defendant is convicted and 0 otherwise. The judge gets utility 1 if they convict when guilty or acquit when innocent, and 0 otherwise. The sender (prosecutor) chooses two distributions

$$\pi(\bullet|\text{guilty}), \quad \pi(\bullet|\text{innocent})$$

over some set of signals S . Assume the judge convicts when they are indifferent.

Consider the following information structures:

- If $|S| = 1$, then no information is sent, and the judge evaluates using the prior and acquits.
- In the fully reveal state,

$$\pi(s_g|\text{guilty}) = 1, \quad \pi(s_i|\text{innocent}) = 1.$$

Then the judge convicts with probability 0.3.

- Finally, consider the following information structure:

$$\pi(s_g|\text{guilty}) = 1, \quad \pi(s_g|\text{innocent}) = 1 - \alpha.$$

The judge's posterior belief can be computed using Bayes' rule:

$$\mathbb{P}(\text{guilty}|s_g) = \frac{\pi(s_g|\text{guilty}) \mathbb{P}(\text{guilty})}{\mathbb{P}(s_g)} = \frac{1 \cdot 0.3}{0.3 \cdot 1 + 0.7 \cdot \alpha} = \frac{3}{3 + 7\alpha}.$$

We want to send s_g as often as possible subject to the judge convicting upon seeing s_g . This motivates setting $\mathbb{P}(\text{guilty}|s_g) = 0.5$ to get $\alpha = 3/7$. Then the judge convicts with probability $0.3 \cdot 1 + 0.7 \cdot (3/7) = 0.6$.



General setup:

- Two agents: sender and receiver.
- Finite set of states Ω .
- Sender and receiver share a common prior $\mu_0 \in \Delta(\Omega)$. Assume $\mu_0(\omega) > 0$ for each $\omega \in \Omega$.
- The receiver has a finite set of actions A .
- Sender and receiver has utility functions

$$u_S : \Omega \times A \rightarrow \mathbb{R},$$

$$u_R : \Omega \times A \rightarrow \mathbb{R}.$$

A **signal structure** is a collection of distributions $\{\pi(\bullet|\omega)\}_{\omega \in \Omega}$, where each $\pi(\bullet|\omega)$ is a distribution over some finite set S of signals.

Given the prior μ_0 and the signal structure π , the receiver forms a posterior belief μ_s given each signal $s \in S$ given by Bayes' rule:

$$\mu_s(\omega) = \frac{\mu_0(\omega) \pi(s|\omega)}{\sum_{\omega'} \mu_0(\omega) \pi(s|\omega')} =: \frac{\mu_0(\omega) \pi(s|\omega)}{\pi(s)}.$$

If the receiver has a belief μ , they will choose $a \in A$ to maximize their expected utility:

$$a \in \arg \max_{a \in A} \sum_{\omega} \mu(\omega) u_R(a, \omega).$$

Assume the receiver breaks ties in favor of the sender. Let $\hat{a}(\omega)$ be this action. This induces expected utility

$$\hat{u}_S(\mu) = \sum_{\omega} \mu(\omega) u_S(\hat{a}(\omega), \omega).$$

Note that we are integrating against the same belief as the receiver, and implicitly assuming that the sender observes the same signal as the receiver.

A signal structure π induces a distribution τ over posteriors, i.e., $\tau \in \Delta(\Delta(\Omega))$. Note that τ has **value**

$$\int \hat{u}_S(\mu) d\tau(\mu).$$

Proposition 11.2. *The expectation of the posterior beliefs is the prior.*

Proof.

$$\sum_s \pi(s) \mu_s(\omega) = \sum_s \pi(s) \frac{\mu_0(\omega) \pi(s|\omega)}{\pi(s)} = \sum_s \mu_0(\omega) \pi(s|\omega) = \mu_0(\omega). \quad \square$$

Given a prior μ_0 , we say $\tau \in \Delta(\Delta(\Omega))$ is **Bayes-plausible** if

$$\int \mu d\tau(\mu) = \mu_0.$$

The previous proposition says that the distribution over posteriors induced by any signal structure must be Bayes-plausible. It turns out that this is the only restriction on the distribution over posteriors:

Proposition 11.3. *There is a signal structure with value v^* if and only if there is a Bayes-plausible distribution over posterior beliefs such that*

$$\int \hat{u}_S(\mu) d\tau(\mu) = v^*.$$

Proof. $\Rightarrow$: by previous proposition.

$\Leftarrow$: Suppose $\tau \in \Delta(\Delta(\Omega))$ is Bayes-plausible and has value v^* . Consider the set $T \subset \mathbb{R}^{|\Omega|+1}$ of all pairs $(\hat{u}_S(\mu), \mu)$ for μ in the support of τ . We know that (v^*, μ_0) is in the convex hull of T . By Carathéodory's theorem (see next theorem), there is a distribution τ^* with finite support $T^0 \subset T$ such that $\sum_{\mu \in T^0} (\hat{u}_S(\mu), \mu) \tau^*(\mu) = (v^*, \mu_0)$. We can turn this into a signal structure by setting $S = T^0$ with

$$\pi(\mu|\theta) = \frac{\mu(\omega) \tau^*(\mu)}{\mu_0(\omega)}.$$

The belief about ω given $\mu \in T^0$ is

$$\frac{\pi(\mu|\omega)\mu_0(\omega)}{\sum_{\omega'} \mu_0(\omega')\pi(\mu|\omega')} = \frac{\mu(\omega)\tau^*(\mu)}{\sum_{\omega'} \mu_0(\omega') \frac{\mu(\omega')\tau^*(\mu)}{\mu_0(\omega')}} = \mu(\omega). \quad \square$$

Theorem 11.4 (Carathéodory's theorem). *Let $X \subset \mathbb{R}^d$ and let x be in the convex hull of X . Then there are $d + 1$ points in X such that x is in the convex hull of these points.*

Now let G denote the set $\{(\hat{u}_S(\mu), \mu)\}_{\mu \in \Delta(\Omega)}$. Let $\text{conv } G$ denote the convex hull of G and define the **concavification** of $\hat{u}_S$ by

$$V(\mu) := \sup\{v : (v, \mu) \in \text{conv } G\}.$$

Note that $V(\mu_0)$ is exactly the optimal utility for the sender.

Thus if $\hat{u}_S$ is concave then $V = \hat{u}_S$ and sender cannot benefit. Conversely if $\hat{u}_S$ is not concave, then there is at least one prior μ at which the sender can benefit from information design.

A Tools

A.1 Convex Geometry

Definition A.1. A **polyhedron** is the solution set to a system of linear inequalities $Ax \leq b$. A **polytope** is a bounded polyhedron. 

Note that a polyhedron is convex.

A.2 Linear Programming

A.2.1 Farkas' Lemma

Lemma A.2 (Farkas). Given $A \in \mathbb{F}^{m \times n}$ and $b \in \mathbb{F}^m$, exactly one of the following holds:

- (i) There exists x such that $Ax \leq b$.
- (ii) There exists $y \geq 0$ such that $yA = 0$ and $yb < 0$.

If $\mathbb{F} = \mathbb{Q}$, then y can be taken to be integers.

Lemma A.3 (Gordan's Lemma). Let A be an $m \times d$ matrix with rational entries. Exactly one of the following holds:

- (i) There exists y such that $Ay \gg 0$.
- (ii) There exists a nonzero vector of nonnegative integers $z \neq 0$ such that $zA = 0$.

Why not both? We will have

$$z(Ay) = (zA)y = 0,$$

but $Ay \gg 0$ and $z \geq 0$, so $z = 0$, a contradiction.

A.3 Minimax Theorem

Consider a two-player zero-sum game with payoff matrix B . Consider mixed strategies $\alpha_1 \in \Delta(A_1)$ and $\alpha_2 \in \Delta(A_2)$ for the two players so that the expected payoff to player 1 is $\alpha_1 B \alpha_2$. The values are

$$V_1^* = \max_{\alpha_1} \min_{\alpha_2} \alpha_1 B \alpha_2$$
$$V_2^* = \min_{\alpha_2} \max_{\alpha_1} \alpha_1 B \alpha_2.$$

We have

Theorem A.4 (Minimax).

$$\max_{\alpha_1} \min_{\alpha_2} \alpha_1 B \alpha_2 = \min_{\alpha_2} \max_{\alpha_1} \alpha_1 B \alpha_2.$$

Proof. Suppose for contradiction that $V_1^* < V_2^*$. This is equivalent to saying that there is no mixed strategy x for player 1 such that

$$xB \geq V_2^* \mathbf{1},$$

□

Two ideas help us write linear systems in standard form:

- To write $Ax \leq b$ in standard form, we can introduce a slack variable s and write $Ax + s = b$ with $s \geq 0$.
- To write $x \in \mathbb{F}$ in standard form, we can write $x = x^+ - x^-$ with x^+, x^- nonnegative.